

ADVANTEST[®]

ADVANTEST CORPORATION

ATL-51
Socket Program
Reference Manual

MANUAL NUMBER 8256293-21

Applicable OS
ASX/U-51

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Table of Contents

Preface	Preface-1
Purpose of This Document	Preface-1
Prerequisites and Target Audiences	Preface-1
Related Manuals	Preface-1
Typographic Convention	Convention-1
Definition of Terms about Parallel Measurement	Convention-1
Description of Test System	Convention-1
Types of Test Head	Convention-1
Reference Indicator	Convention-2
1. Overview	1-1
1. 1 Roles of a Socket Program	1-1
1. 1. 1 Definition Related to Pin Connection	1-1
1. 1. 2 Definition of Converted Symbols	1-1
1. 1. 3 Definition of Parallel Test	1-1
2. Format of Socket Program	2-1
2. 1 Configuration of Socket Program	2-1
2. 1. 1 Configuration of Socket Program	2-1
2. 1. 2 File Name	2-2
2. 1. 3 DR Pin MAX, CP Pin MAX	2-2
2. 2 Definition of DUT Pin Name	2-4
2. 2. 1 DUT Pin Name	2-4
2. 2. 2 DUT Pin Number	2-6
2. 2. 3 Tester Channel Number	2-7
2. 3 Definition of Converted Symbols	2-9
2. 3. 1 Format	2-9
2. 3. 2 Symbol Name	2-9
2. 4 Definition of VS9-12	2-10
2. 4. 1 Format	2-10

2. 5	Definition of Parallel Test	2-12
2. 5. 1	Format	2-12
2. 6	Addition of Revision Number to Object File at Compilation	2-25
2. 6. 1	%REMARK Statement	2-25

3. Allocation of Tester Channels and DC Measuring Units

3-1

3. 1	Tester Channel Allocation Table	3-1
3. 2	DC Measuring Unit Number	3-3

4. 1600(576IO)CH or 800(288IO)CH Test Head

4-1

4. 1	Test Head Configurations and Available Child Pins	4-1
4. 2	DC Dedicated Channels and Option Driver Channels	4-2
4. 3	Tester Channel Allocation Table (1)	4-3
4. 4	Tester Channel Allocation Table (2)	4-4
4. 5	Tester Channel Allocation Table (3)	4-5
4. 6	Tester Channel Allocation Table (4)	4-7
4. 7	Tester Channel Allocation Table (5)	4-9
4. 8	Tester Channel Allocation Table (6)	4-10
4. 9	Tester Channel Allocation Table (7)	4-11
4.10	DC Measuring Unit Allocation Table	4-13

5. 1600(288IO)CH or 800(144IO)CH Test Head

5-1

5. 1	Test Head Configurations and Available Child Pins	5-1
5. 2	DC Dedicated Channels and Option Driver Channels	5-2
5. 3	Tester Channel Allocation Table (1)	5-3
5. 4	Tester Channel Allocation Table (2)	5-4
5. 5	Tester Channel Allocation Table (3)	5-5
5. 6	Tester Channel Allocation Table (4)	5-6
5. 7	DC Measuring Unit Allocation Table	5-7

6. 1664(576IO)CH or 1664(640IO)CH Test Head	6-1
6. 1 Test Head Configurations and Available Child Pins	6-2
6. 2 DC Dedicated Channels and Driver Dedicated Channels	6-4
6. 3 Tester Channel Allocation Table (1)	6-5
6. 4 Tester Channel Allocation Table (2)	6-6
6. 5 Tester Channel Allocation Table (3)	6-7
6. 6 Tester Channel Allocation Table (4)	6-8
6. 7 Tester Channel Allocation Table (5)	6-9
6. 8 Tester Channel Allocation Table (6)	6-10
6. 9 Tester Channel Allocation Table (7)	6-11
6.10 Tester Channel Allocation Table (8)	6-13
6.11 DC Measuring Unit Allocation Table	6-15
 7. 352CH Test Head	 7-1
7. 1 Test Head Configurations and Available Tester Channels	7-1
7. 2 Tester Channel Allocation Table of T5591	7-2
7. 2. 1 Tester Channel Allocation Table (1)	7-2
7. 2. 2 Tester Channel Allocation Table (2)	7-3
7. 2. 3 Tester Channel Allocation Table (3)	7-5
7. 2. 4 Tester Channel Allocation Table (4)	7-7
7. 2. 5 Tester Channel Allocation Table (5)	7-9
7. 3 Tester Channel Allocation Table of T5591R	7-12
7. 3. 1 Tester Channel Allocation Table (1)	7-12
7. 3. 2 Tester Channel Allocation Table (2)	7-14
7. 4 DC Measuring Unit Allocation Table	7-17
 8. 2880CH Test Head	 8-1
8. 1 Test Head Configurations and Available Child Pins	8-1
8. 2 DC Dedicated Channels and Driver Dedicated Channels	8-2
8. 3 Tester Channel Allocation Table (1)	8-3
8. 4 Tester Channel Allocation Table (2)	8-4
8. 5 Tester Channel Allocation Table (3)	8-6

8. 6	Tester Channel Allocation Table (4)	8-8
8. 7	Tester Channel Allocation Table (5)	8-10
8. 8	DC Measuring Unit Allocation Table	8-12
9.	1344CH Test Head	9-1
9. 1	Test Head Configurations, Available Child Pins, and Channel Numbers	9-1
9. 2	Tester Channel Allocation Table (1)	9-2
9. 3	Tester Channel Allocation Table (2)	9-3
9. 4	Tester Channel Allocation Table (3)	9-4
9. 5	Tester Channel Allocation Table (4)	9-6
9. 6	DC Measuring Unit Allocation Table	9-7
10.	3712(1280IO)CH or 3456(1280IO)CH Test Head	10-1
10. 1	Test Head Configurations and Available Child Pins	10-1
10. 2	DC Dedicated Channels and Level Driver Channels	10-2
10. 3	Tester Channel Allocation Table (1)	10-3
10. 4	Tester Channel Allocation Table (2)	10-4
10. 5	Tester Channel Allocation Table (3)	10-5
10. 6	Tester Channel Allocation Table (4)	10-6
10. 7	Tester Channel Allocation Table (5)	10-7
10. 8	Tester Channel Allocation Table (6)	10-8
10. 9	Tester Channel Allocation Table (7)	10-9
10.10	Tester Channel Allocation Table (8)	10-10
10.11	Tester Channel Allocation Table (9)	10-11
10.12	Tester Channel Allocation Table (10)	10-13
10.13	Tester Channel Allocation Table (11)	10-15
10.14	Tester Channel Allocation Table (12)	10-17
10.15	Tester Channel Allocation Table (13)	10-19
10.16	Tester Channel Allocation Table (14)	10-21
10.17	Tester Channel Allocation Table (15)	10-23

10.18	Tester Channel Allocation Table (16)	10-24
10.19	Tester Channel Allocation Table (17)	10-25
10.20	Tester Channel Allocation Table (18)	10-26
10.21	DC Measuring Unit Allocation Table	10-27

11. 3072CH Test Head 11-1

11. 1	Test Head Configurations, Available Child Pins, and Channel Numbers	11-1
11. 2	Tester Channel Allocation Table (1)	11-2
11. 3	Tester Channel Allocation Table (2)	11-3
11. 4	Tester Channel Allocation Table (3)	11-4
11. 5	Tester Channel Allocation Table (4)	11-5
11. 6	Tester Channel Allocation Table (5)	11-6
11. 7	Tester Channel Allocation Table (6)	11-7
11. 8	Tester Channel Allocation Table (7)	11-9
11. 9	Tester Channel Allocation Table (8)	11-11
11.10	Tester Channel Allocation Table (9)	11-13
11.11	DC Measuring Unit Allocation Table	11-14

12. 3072CH Test Head with High-speed Clock Drivers 12-1

12. 1	Test Head Configurations, Available Child Pins, and Channel Numbers	12-1
12. 2	High-speed Clock Driver Channel	12-2
12. 3	Tester Channel Allocation Table (1)	12-3
12. 4	Tester Channel Allocation Table (2)	12-4
12. 5	Tester Channel Allocation Table (3)	12-5
12. 6	Tester Channel Allocation Table (4)	12-6
12. 7	Tester Channel Allocation Table (5)	12-7
12. 8	Tester Channel Allocation Table (6)	12-9
12. 9	Tester Channel Allocation Table (7)	12-11

12.10	Tester Channel Allocation Table (8)	12-13
12.11	DC Measuring Unit Allocation Table	12-15

**Appendix 1.Compilation Messages, Compilation
Error Codes** A1-1

A1. 1	Compilation Messages	A1-1
A1. 2	Compilation Error Codes	A1-2

Appendix 2.Example of Socket Program A2-1

A2. 1	Example of Socket Program Description	A2-1
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List of Figures F-1

List of Tables T-1

Index I-1

Preface

Purpose of This Document

This document explains how to describe the socket program required to perform the simultaneous measurement of multiple devices, and shows tester channel allocation table.

Prerequisites and Target Audiences

This document is created on the assumption that the audiences have experienced following training courses, or have the equivalent knowledge and experiences:

- T5500 memory test systems primary course
- T5360, T5380, T5330 and T3330 memory test systems primary course

The target audiences are as follow:

- Those who related to the creation of ATL programs

Related Manuals

ATL-51 Test Plan Program Reference Manual (No. : 8431498)

This manual explains how to create the test plan programs on T5500 series memory test systems.

ATL-51 Pattern Program Reference Manual (T5500) (No. : 8256292)

This manual explains how to create the pattern program on T5500 series memory test systems and outline of the algorithmic pattern generator (ALPG) which is used to generate test pattern.

ATL-51 Pattern Program Reference Manual (T5376/T5375/T5371/T5336/T5311) (No. : 8289058)

This manual explains how to create the pattern program on T5376, T5375, T5371, T5336 and T5311 series memory test systems and outline of the algorithmic pattern generator (ALPG) which is used to generate test pattern.

ATL-51 Scramble Program Reference Manual (No. : 8256294)

This manual explains scramble program which is an address conversion program for memory device measurement.

T5500 Series Tester Utility Program Reference Manual (No. : 8256295)

This manual explains useful tools (utility programs) to use testers.

T5500 Series Controller Commands Reference Manual (No. : 8256297)

This manual explains program operations activated by controller command.

ATLworks Tool User's Manual (No. : 8311174)

This manual explains the ATLworks, a software, which is used for the purpose of reducing the test program debugging time and improving device evaluation efficiency.

ASX/U-51 Hybrid Controller User's Manual (No. : 8409598)

This manual explains the hybrid controller which makes C language and C++ language possible to be used as a programming language for the device measurement programs in addition to the ATL language.

ASX/U-51 Hybrid Controller Reference Manual (No. : 8409600)

This manual explains utility programs which is useful for using with library functions of a hybrid controller or a hybrid controller itself.

Error Code Table Reference Manual (No. : 8141091)

This manual explains various errors during the compilation of program execution.

XATL/U-51 Cross Compiler User's Manual (No. : 8256237)

This manual explains the XATL-U-51 cross compiler, a software, which is used to compile ATL test programs into object programs under the SunOS.

ADVAnsite Test System Software Installation/System Start User's Manual VOL1 Basic OS Installation (No. : 8227728)

This manual explains how to install the ASX/U system software. ASX/U system software consists of a basic OS (Solaris) and a tester environment software. This manual explains how to install the basic OS (Solaris).

ADVAnsite Test System Software Installation/System Start User's Manual VOL2 Tester Environment Software Installation (No. : 8227729)

This manual explains how to use the ASX/U system software, covering from installation of the software to starting of the test system. ASX/U system software consists of a basic OS (Solaris) and a tester environment software. This manual also explains how to install the tester environment software and start the test system.

T5500 Series Memory Test System Control Box/Trigger Panel User's Manual (No. : 8256298)

This manual explains control box/trigger panel unit which is used to control operations of device program execution and debugging in the controller mode (tester driver).

T5300 Series Flash Memory Parallel Measurement Function Description (No. : 8239605)

This manual explains the flash memory parallel measurement function of the T5300 series memory test systems and how to generate a program to use this function.

T5500 Series Memory Test System Calibration Program Reference Manual (No. : 8256296)

This manual explains the calibration program which is used to calibrate the various tester timings based on the conditions set in a test program generated by a user.

T5500 Series Memory Test System Precision Calibration Program Reference Manual (No. : 8311137)

This manual explains the high-precision calibration program which is used to calibrate the various tester timings based on the conditions set in a test program generated by a user.

T5500 Series Memory Test System Precision Calibration2 Program Reference Manual (No. : 8350393)

This manual explains the high-precision calibration program which is used to calibrate the various tester timings based on the conditions set in a test program generated by a user.

Typographic Convention

Definition of Terms about Parallel Measurement

Terms about parallel measurement used in this manual are defined as follows.

- Parallel measurement of multiple devices
It is generally used for parallel measurement. It actually means the multiple devices (except 1STN on 1DUT) measurement declared by using DEFINE MULTIPLE statement in the socket program. Parallel measurement means the measurement described above.
- Parallel measurement of multiple devices on n-station
It is used when the number of stations (n=1,2) needs to be defined for the parallel measurement.
- Single device measurement in the parallel measurement of multiple devices mode
It is used when the parallel measurement mode is declared by DEFINE MULTIPLE statement in the socket program, and also the measured device and station are both one (1STN on 1DUT). It is used when the measurement needs to be distinguish from parallel measurement of multiple devices.
- Single device measurement
It is used for a measurement which is not parallel measurement. It actually means the measurement without using the socket program, or without using the DEFINE MULTIPLE statement in the socket program.

Description of Test System

Unless otherwise specifically noted, descriptions of test system in this manual are applied to models below.

Table C-1 Applicable Models to Descriptions of Test System

Description of test system	Applicable models
T5500	T5593, T5592, T5591, T5591R, T5586, T5585, T5581, T5581H, T5581P and T5571P

Types of Test Head

Following test heads are available for test systems.

Table C-2 Test Head Corresponding to Test System

T5593	3072CH test head
T5592	1344CH test head
T5591 T5591R	352CH test head
T5586 T5585	2880CH test head
T5581 T5581H	1600(576IO)CH test head
	1600(288IO)CH test head
T5581P T5571P	800(288IO)CH test head
	800(144IO)CH test head
T5376	3456(1280IO)CH test head
T5377 T5375	3712(1280IO)CH test head
T5371	1664(640IO)CH test head
T5336 T5311	1664(576IO)CH test head

Reference Indicator

Reference indicators use double-quote (") symbols.

Table C-3 Reference Indicator

Typeface or symbol	Explanation	Example
" "	Indicates the chapter titles and section titles of a manual to refer to.	For more information, refer to 2. "Restarting and Quitting."
<i>Italic</i>	Indicates the title of a manual to refer to.	For more information, refer to <i>Installation Guide</i> .

1. Overview

The following three types of definitions can be described in the socket program used in ASX/U-51.

- [Definition Related to Pin Connection](#)
- [Definition of Converted Symbols](#)
- [Definition of Parallel Test](#)

Test plan programs are compiled based on the definition described in the socket program. Therefore, when the socket program is used, the socket program must be compiled before test plan programs are compiled.

! Important

When the socket program is recompiled after being changed, test plan programs must also be recompiled.

1. 1 Roles of a Socket Program

1. 1. 1 Definition Related to Pin Connection

This definition defines the relation between the tester channel number and the pin number of the device under test (DUT) on the performance board. The information defined here is included in the object file when test plan programs are compiled, and used to relate the DUT pin number and the tester channel number. This definition enables programming focusing on the DUT pin number instead of the tester channel number, and makes the program easier to understand.

1. 1. 2 Definition of Converted Symbols

This definition enables the registered characters (i.e., VS1, IN1, etc.) used in test plan programs to be described by other specified characters (i.e., VCC, VIN, etc.).

1. 1. 3 Definition of Parallel Test

This definition relates the tester channel number and the DUT pin number for parallel test of multiple number of items. If you use the socket program in which this definition is described, you can change the test plan program for measurement of single number of unit to the test plan program for parallel test of multiple items.

2. *Format of Socket Program*

This chapter describes the following contents related to the format of socket program.

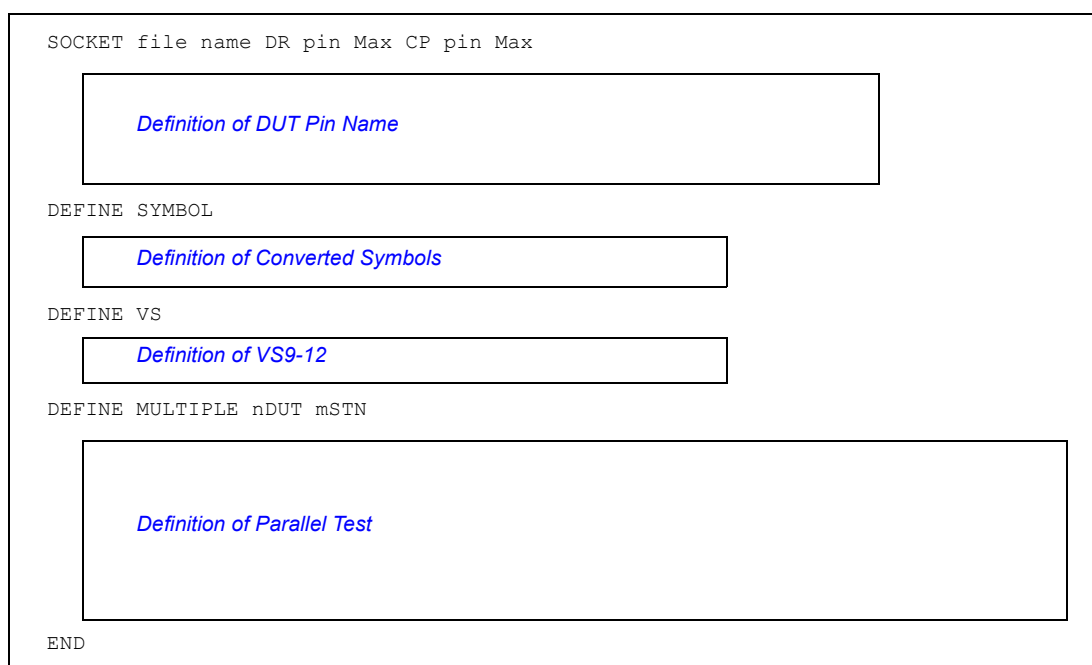
- [Configuration of Socket Program](#)
- [Definition of DUT Pin Name](#)
- [Definition of Converted Symbols](#)
- [Definition of VS9-12](#)
- [Definition of Parallel Test](#)

2. 1 Configuration of Socket Program

2. 1. 1 Configuration of Socket Program

A socket program starts with the SOCKET statement and ends with the END statement. A socket program consists of the following four definition groups described in the following orders:

Figure 2-1 Configuration of Socket Program



 Tip

Only necessary definitions, not necessarily all definitions, need to be described.

Special characters**; (semicolon)**

During the compilation, one character string after the semicolon (;) before the CR is ignored.

@ (at sign)

During the compilation, one character after the at sign is ignored. This sign is used for when a long statement cannot be written within a line and needs to be divided into two lines. If you write this sign at the end of first line, it is compiled as one statement for two lines. @ sign can be used for a line starting with (;).

2. 1. 2 File Name

The file name specified here becomes the object file name to be created at the compilation. A file name must be specified by using up to 32 characters which include alphanumeric characters, periods (.), and dollar signs (\$). "soc" is attached to the object file name as file identifier.

 Important

When using the socket program, write the same file name as the one specified here on the first line of the test plan program.

2. 1. 3 DR Pin MAX, CP Pin MAX

Specify the maximum pin numbers of the driver channel (DR pin) and the comparator channel (CP pin) of the tester connected to DUT as DR pin MAX and CP pin MAX in decimal number, respectively.

◆ When 15 and 40 are Specified as DR Pin MAX and CP Pin MAX, Respectively

```
SOCKET MUT1K 15 40
```

The available maximum pin number

In both cases, the available maximum pin number is 512.

Default

The default value is 256, however, if one of the above values is specified, the specified value is used for both numbers.

 Tip

This specification is conducted to optimize the size of the object file to be created. Even if omitted, it has no effect on operation.

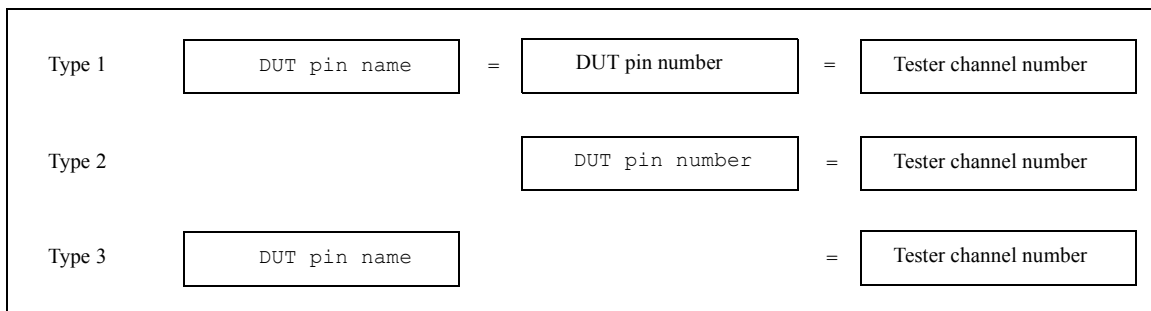
When the MAX pin number is not described for the DUT pin number, the next tester channel number is automatically allocated to the undescribed DUT pin number.

- 1. If the tester channel with the same number as the DUT pin number is not specified, the tester channel number is allocated.*
 - 2. For the DUT pin numbers to which no numbers are allocated in the above 1, unused tester channel numbers are allocated from the smaller ones sequentially.*
-

2. 2 Definition of DUT Pin Name

This definition enables description of test plan programs using DUT pin names (i.e., WE, OE A0, A1, etc.) or DUT pin numbers, instead of tester channel numbers. There are following three types of definition:

Figure 2-2 Definition of DUT Pin Name



◆ Specification Example of DUT Pin Names of MA0, MA1, MA2, ..., MA9

```

MA0 = PD2 = 1
MA1 = PD3 = 6
MA2 = PD4 = 7
MA3 = PD5 = 8
MA4 = PD6 = 9
MA5 = PD7 = 10
MA6 = PD9 = 5
MA7 = PD10 = 4
MA8 = PD11 = 3
MA9 = PD12 = 2

```



Tip

This definition can be omitted when DUT pin names and DUT pin numbers are not used in the test plan programs.

2. 2. 1 DUT Pin Name

A DUT pin name must be specified by using up to 32 alphanumeric characters which include alphanumeric characters and underscores (_), and which start with an alphabetic character.



Limitation

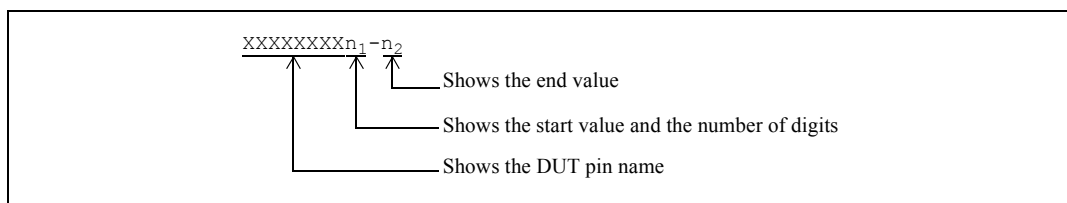
- Do not use PD, PC, or P as a pin name.

- A registered string of a test plan program and a mnemonic of a statement (excluding pin specification) cannot be specified as a DUT pin name.

◆ Consecutive Specification of DUT Pin Names

When a sequential serial number is attached to the end of a DUT pin name, by adding a hyphen (-) to the end of the name, multiple DUT pin names can be specified by one line. DUT pin names can be specified either in ascending or descending order.

Figure 2-3 Consecutive Specification of DUT Pin Names



◆ Specification Example of DUT Pin Names of MA0, MA1, MA2, ..., MA9 by One Line

MA0-9 = PD2-7, PD9-12 = 1, 6-10, 5-2

The above example has the same meaning as the following descriptions:

```
MA0 = PD2 = 1
MA1 = PD3 = 6
MA2 = PD4 = 7
MA3 = PD5 = 8
MA4 = PD6 = 9
MA5 = PD7 = 10
MA6 = PD9 = 5
MA7 = PD10 = 4
MA8 = PD11 = 3
MA9 = PD12 = 2
```

! Important

When making consecutive specification, any number cannot be specified in a DUT pin name except for a sequential serial number attached to its end.

A3XB120 ← Cannot specify

Only one list of DUT pin names can be specified in one line.

MA0-1, MA2-9 = ... ← Cannot specify

Pin name + Number - Number
 └──────────────────┘
 This unit is called "one list".

◆ When the Sequential Serial Number at the End of a DUT Pin Name Starts with 0

When the sequential serial number at the end of a DUT pin name starts with 0, divide the description of consecutive specification into two lines as follows:

◆ Specification Example of DUT Pin Names of MA0, MA1, MA2, ..., MA9 by One Line (in Ascending Order)

```
MA00-09 = ...
MA10-11 = ...
```



Tip

Defining "MA00-11 = " develops MA00, MA01, MA02, ..., MA09, MA010, MA011.

◆ Specification Example of DUT Pin Names of MA0, MA1, MA2, ..., MA9 by One Line (in Descending Order)

```
MA11-10 = ...
MA09-00 = ...
```



Tip

Defining "MA10-01 = " develops MA10, MA9, MA8, ..., MA3, MA2, MA1.

2. 2. 2 DUT Pin Number

DUT pin numbers are defined when test plan programs are described by pin numbers of DUT. Specify the DUT pin number by attaching PD, PC, or P to the head of the pin number. The meanings of each symbol are as follows:

PD

DUT input pin

PC
DUT output pin

P
DUT I/O (common input/output) pin

Available format

Table 2-1 Available Formats of DUT Pin Numbers

Format	Meaning
PDn	Input pin for pin number n
PCn	Output pin for pin number n
Pn	I/O pin for pin number n
PDn ₁ -n ₂	Input pin for pin number n ₁ to n ₂
PCn ₁ -n ₂	Output pin for pin number n ₁ to n ₂
Pn ₁ -n ₂	I/O pin for pin number n ₁ to n ₂

Available values for n, n₁, and n₂
1 ≤ n, n₁, n₂ ≤ Maximum pin number
(provided that n₁ ≠ n₂)
When making consecutive specification, up to 20 lists of DUT pin names can be specified in one line.

◆ Description Example of Two Lists for DUT Pin Names in One Line

MA0-9 = PD2-7, PD9-12 = ...

The above example has the same meaning as the following descriptions:

MA0 = PD2 = ...
MA1 = PD3 = ...
...
MA5 = PD7 = ...
MA6 = PD9 = ...
...
MA9 = PD12 = ...

2. 2. 3 Tester Channel Number

Specify the tester channel number on the performance board.

Table 2-2 Available Format

Format	Meaning
n	Tester channel number n
n ₁ -n ₂	Tester channel number n ₁ to n ₂

**Tip**

PD, PC, or P is not attached to the tester channel number.

Available values for n, n₁, and n₂

In case of driver channel

$$1 \leq n, n_1, n_2 \leq 512$$

In case of comparator channel

$$33 \leq n, n_1, n_2 \leq 512$$

In case of I/O channel

$$1 \leq n, n_1, n_2 \leq 512$$

**Tip**

In any of the above cases, n₁ ≠ n₂. If the same tester channel numbers are written redundantly, an error occurs.

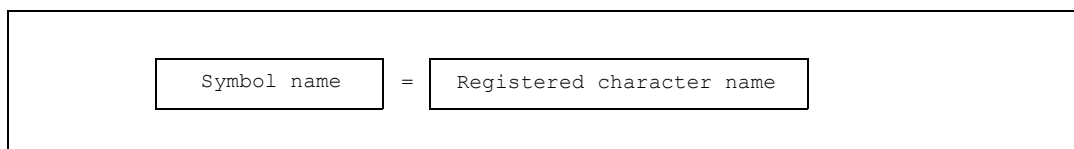
2. 3 Definition of Converted Symbols

This definition is used to replace the registered characters (i.e., VS1, IN1, etc.) used in test plan programs with other specified names (symbol names). This definition can be omitted when symbol names are not used.

2. 3. 1 Format

A symbol name is defined in the following format. A symbol name and a registered character name can be specified in one line on a one-on-one basis. Consecutive specification of multiple symbol names or registered character names in one line is not possible.

Figure 2-4 Definition of Converted Symbols



2. 3. 2 Symbol Name

A symbol name must be specified by using up to 32 alphanumeric characters which include alphanumeric characters and underscores (_), and which start with an alphabetic character.

◆ Examples of Definition for Converted Symbols

(Definition in the socket program)

```

DEFINE SYMBOL
  VIN  = IN
  VOUT = OUT
  
```

(Definition in the test plan program)

```

VIN(1) = 5V, 2V
VOUT(1) = 4V, 1V
  
```

The above specification has the same meaning as one specified followings in the test plan program:

```

IN1      = 5V, 2V
OUT1     = 4V, 1V
  
```

In the following examples, the registered character names "IN" and "OUT" are defined as "VIN" and "VOUT", respectively, according to the definition of converted symbols. When compiled, "VIN" and "VOUT" in test plan programs are replaced with "IN" and "OUT", respectively.

2. 4 Definition of VS9-12

This definition is used to allocate a hardware channel number of PPS^{*1} to the registered character name VS9, VS10, VS11, or VS12.

Available systems

T5593, T5592, T5591, T5591R, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336 and T5311



Tip

*This definition is used when connecting multiple PPSs to one DUT or paradriving^{*2} PPSs.*

**1 Device power supply. Abbreviation for Programmable Power Supply.*

**2 Supply a huge electric current for DUT by connecting multiple PPSs in parallel.*

2. 4. 1 Format

VS9-12 is defined in the following format.

$$VS_n = PPS_{m_1-m_2}$$

n

Specification of the channel number of a VS statement.

Available values

n = 9 to 12

m₁-m₂

Specification of hardware channel number for PPS. For defining two or more PPSs, parallel drive specification is used.

Available values

1 to 320

Available parallel drives

1 to 256

Default

When the above specifications are omitted, the following specifications are used:

```
DEFINE VS
VS9  = PPS97
VS10 = PPS98
VS11 = PPS99
VS12 = PPS100
```

 Important

When PPS97 to PPS100 are defined for other VS_n(_n = 1 to 12), VS_n(_n = 9 to 12) that has no PPS available as default value cannot be specified.

◆ **Example in which PPS97, 98, and 99 are Allocated to VS9, and PPS33 is Allocated to VS10**

```
DEFINE VS
VS9  = PPS97-99
VS10 = PPS33
```

PPS97, 98, and 99 are allocated to VS9, and PPS33 is allocated to VS10. In this case, because PPS99 is already allocated to VS9, PPS99 can no longer be used for VS11. There is no description about VS12, however, PPS100 can be used as VS12.

2. 5 Definition of Parallel Test

This definition is used to test multiple DUTs simultaneously. When this definition is omitted, single measurement is used.

2. 5. 1 Format

The format of definition for parallel test is as follows:

Figure 2-5 Format of Definition for Parallel Test

```

DEFINE MULTIPLE kDUT mSTN ..... (1)
  BIT = a ..... (2)
  PPS = b ..... (3)
  PIN MODE = x ..... (4)
  CHTYPE n ..... (5)
  HANDLER (or PROBER) ..... (6)
  HVPPSn1=HVPPSm1, HVPPSm2, HVPPSm3, HVPPSm4 ..... (7)
  ...
  HVPPSn4=HVPPSm1, HVPPSm2, HVPPSm3, HVPPSm4
  PARALLEL VSn=m ..... (8)
  SEPARATE PPS m ..... (9)
  SEPARATE DR m ..... (10)
  DUT# = VSn1, PDn2, PCn3, Pn4 ..... (11)
  DUT1 = PPSm1, PDm2A1, PCm3A, Pn4A
  ...
  DUTk = PPSk1, PDk2x1, PCk3x, Pn4x ..... (12)

```

Explanations of number (1) to (12) in the format are as follows:

(1) DEFINE MULTIPLE kDUT mSTN

Declaration of definition for parallel test.

k

Specification of the number of DUTs to be tested in one station.

Available values

k = 1, 2, 4, 8, 16, 32, 64, 128, 256

m

Specification of the number of stations to be used.

Available values

m = 1, 2

(2) BIT = a

Specification of the bit configuration for DUT.

Available values

The values available for a differ depending on CHTYPE_n as shown in the table below.

Table 2-3 CHTYPE_n and Available Bit Numbers

CHTYPE _n	a
CHTYPE3	4, 8, 9, 11, 16, 18, 22, 32, 36, 44, 64, 72, 88, 128, 144, 176
CHTYPE4 or 5	1, 4, 8, 9, 16, 18, 19 to 72
CHTYPE6	1, 4, 8, 9, 10, 16, 18, 19 to 36, 40
CHTYPE9	1, 4, 8, 9, 16, 18, 32, 36, 64, 72
CHTYPE10	1, 4, 8, 9, 16, 18, 24, 32, 36, 48, 64, 72, 96
CHTYPE11	1, 4, 5, 8, 9, 10, 16, 18, 20, 32, 36, 40
CHTYPE12 or 13	1, 4, 8, 9, 12, 16, 18, 24, 32, 36, 48, 64, 72, 96

(3) PPS = b

Specification of the number of PPSs used in one DUT.

Available values

b = 1 to 8



VS=b can also be defined.

(4) PIN MODE = x

Specification of pin mode. This specifies which tester channel is allocated to what number of DUT.

Available values

x = A, B, C

➔ For more information on pin mode A, B and C, refer to "Tester channels allocation table".

(5) CHTYPE_n

Specification of the channel number type of test head to be used.

Table 2-4 CHTYPE_n and Channel Number Type of Test Head

CHTYPE _n	Channel Number Type of Test Head
CHTYPE3	352CH test head
CHTYPE4	1600(576IO)CH/800(288IO)CH test head
CHTYPE5	1600(288IO)CH/800(144IO)CH test head
CHTYPE6	1664(576IO)CH/1664(640IO)CH test head

CHTYPE _n	Channel Number Type of Test Head
CHTYPE9	2880CH test head
CHTYPE10	1344CH test head
CHTYPE11	3712(1280IO)CH test head
	3456(1280IO)CH test head
CHTYPE12	3072CH test head
CHTYPE13	3072CH test head with high-speed clock driver

Default

CHTYPE4

(6) HANDLER (or PROBER)

Specification to use the handler (or prober). Specify either one of HANDLER or PROBER. This specification enables communication through GPIB interface between the test system and handler (or prober).

Default

By default, manual station is specified.

 Important

To enable communication through GPIB interface, necessary settings of /STHAND and /PROBSEL utility as well as this specification need to be completed.

➔ For more information on /STHAND or /PROBSEL utilities, refer to [T5500 Series Tester Utility Program Reference Manual \(No. : 8256295\)](#).

(7) HVPPSn=HVPPSm₁, HVPPSm₂, HVPPSm₃, HVPPSm₄

Specification of 16V range VS channel (VS9-12) to be set simultaneously.

HVPPSn

Specification of HVPPS channels (HVPPSm) to be set simultaneously. Up to 4 HVPPSm (m = 9, 10, 11, 12) can be allocated to one HVPPSn.

Available values

n = 9, 10, 11, 12

 Important

- n value must be specified in ascending order.
- HVPPSn cannot be overloaded.

HVPPSm

Specification of HVPPS channels to be set simultaneously.

Available values

m = 9, 10, 11, 12

! Important

HVPPSm cannot be overloaded.

Default

```

HVPPS9  = HVPPS9
HVPPS10 = HVPPS10
HVPPS11 = HVPPS11
HVPPS12 = HVPPS12

```

◆ Specification Example of HVPPS Statement

By the following specifications, the condition of VS9 statement written in the test plan program is set to VS9 and VS11 simultaneously. The condition of VS10 statement is set to VS10 and VS12 simultaneously.

```

HVPPS9 = HVPPS9, HVPPS11
HVPPS10 = HVPPS10, HVPPS12

```

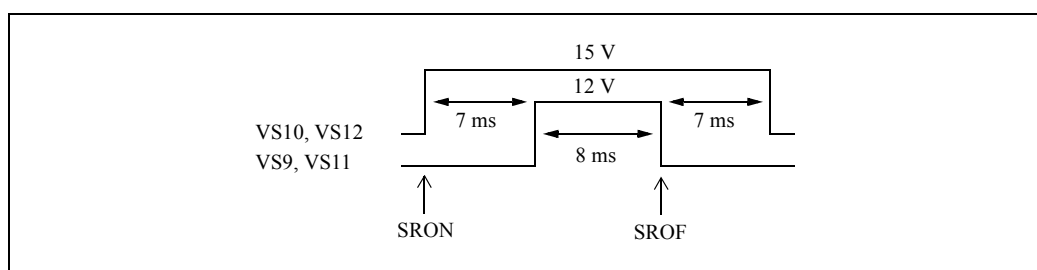
◆ Specification Example of Test Plan Programs

```

VS9=12V,R(16V)    ; Specifies 12V to VS9 and VS11
VS10=15V,R(16V)   ; Specifies 15V to VS10 and VS12
...
TIME1=7MS :VS10   ;
TIME2=8MS :VS9    ;
SRON                      ;
                        ;
SROF                      ;
PCON=VS9             ; Connects a bypass capacitor to VS9 and VS11
VCON=VS10            ; Opens a relay of VS10 and VS12

```

Figure 2-6 Changes in the Voltage of VS9, VS10, VS11 and VS12



! Important

Settings to VS9 to VS12 by the following ATL statements can be performed simultaneously by the HVPPS statement.

```
VS statement
PCON statement
VCON statement
RON/ROF statement
SRON/SROF statement
TIME statement
```

Settings to VS9 to VS12 by the following ATL statements written in the BURST BLOCK statement can be performed simultaneously by the HVPPS statement. Note that only the specified channels, not the VS statements written in the BURST BLOCK statement, are to be set simultaneously.

```
BURST BLOCK BEGIN
BURST RON/ROF statement
PCON statement
VCON statement
BURST BLOCK END
```

The MEAS VS statements (or START VS statements) are not set simultaneously by the HVPPS statement, therefore, to perform parallel test, write all necessary VS channels as follows:

```
LIMIT VS9 = ...
LIMIT VS10 = ...
LIMIT VS11 = ...
LIMIT VS12 = ...
MEAS VS9, VS10, VS11, VS12
```

Tip

HVPPS can also be described as VS.

(8) PARALLEL VS_n=m

Specification of PPS parallel drive. Specify the number of PPSs to be performed the parallel drive by VS_n. By this specification, test conditions of multiple PPSs connected in parallel to one DUT can be specified as if they were one VS_n.

Available values

n = 1 to 8

m = 1 to 256

Default

```

PARALLEL VS1 = 1
PARALLEL VS2 = 1
PARALLEL VS3 = 1
PARALLEL VS4 = 1
PARALLEL VS5 = 1
PARALLEL VS6 = 1
PARALLEL VS7 = 1
PARALLEL VS8 = 1

```

◆ Description Example of PARALLEL VS Statement

```

SOCKET SA0404S1
  DEFINE MULTIPLE 4DUT 1STN
    BIT=4
    PPS=2*1
    PIN MODE=A
    CHTYPE4
    PARALLEL VS1=4
    PARALLEL VS2=2
DUT#  = VS1      VS2      PD1-32   PD65-68   PD77-80   PD93-94   P33-36
DUT1  = PPS1-4   PPS17,PPS18 PD1-32A1 PD65-68A1 PD77-80A1 PD93-94A1 P33-36A
DUT2  = PPS5-8   PPS19,PPS20 PD1-32B1 PD65-68B1 PD77-80B1 PD93-94B1 P33-36B
DUT3  = PPS9-12  PPS21,PPS22 PD1-32C1 PD65-68C1 PD77-80C1 PD93-94C1 P33-36C
DUT4  = PPS13-16 PPS23,PPS24 PD1-32D1 PD65-68D1 PD77-80D1 PD93-94D1 P33-36D
END

```

**1 By specifying PPS = 2, VS1 and VS2 can be specified to the DUT# line.*

◆ Description Example of Test Plan Programs

```

PRO MSAMPLE2 SA0404S1
  VS1=5V,R10V,M400MA      ; Specifies 5V to PPS1-16
  VS2=7V,R10V,M400MA      ; Specifies 7V to PPS17-24
  MEAS VS1                 ; Total sum of current value for PPS1-4 is VS1
                           ; measurement value for DUT1
      :                   ; Total sum of current value for PPS5-8 is VS1
                           ; measurement value for DUT2
      :                   ; Total sum of current value for PPS9-12 is VS1
                           ; measurement value for DUT3
      :                   ; Total sum of current value for PPS13-16 is VS1
                           ; measurement value for DUT4
  MEAS VS2                 ; Total sum of current value for PPS17-18 is VS2
                           ; measurement value for DUT1
      :                   ; Total sum of current value for PPS19-20 is VS2
                           ; measurement value for DUT2
      :                   ; Total sum of current value for PPS21-22 is VS2
                           ; measurement value for DUT3
      :                   ; Total sum of current value for PPS23-24 is VS2
                           ; measurement value for DUT4
END

```

— ! Important —

Settings to VSn by the following ATL statements can be performed by parallel drive specification.

```

VS statement
PCON statement
VCON statement
RON/ROF statement
SRON/SROF statement
TIME statement
MEAS/START VS statement
SET REJECTION statement
SET/RESET EXCLUSION2 statement
MULTIPLE DUT statement

```


Settings to VS_n by the following ATL statements written in the BURST BLOCK statement can be performed simultaneously by the parallel drive specification.

```
BURST BLOCK BEGIN
    BURST RON/ROF statement
    PCON statement
    VCON statement
    VS statement
BURST BLOCK END
```

(9) SEPARATE PPS m

Specification to use one PPS for multiple DUTs by branching the PPS. Specify the number of DUTs into which one PPS is branched to m. By this specification, the same PPS number can be written up to m times by using the specification of (12). However, the same PPS number cannot be specified for the different VSs in the same DUT. For example, PPS1 cannot be specified both for VS1 and VS2. In this case, a compilation error occurs.

Available values

m = 1, 2, 4

Default

```
SEPARATE PPS 1
```

◆ Example of a Correct Description

```
SEPARATE PPS 2
DUT# = VS1, ...
DUT1 = PPS1, ...
DUT2 = PPS1, ...
```

◆ Example of an Incorrect Description

```
SEPARATE PPS 2
DUT# = VS1, VS2, PD1-10, ...
DUT1 = PPS1, PPS1*1, PD1-10A1, ...
```

**1 Cannot specify.*

(10) SEPARATE DR m

Specification to use one driver channel for multiple DUTs by branching the driver. Specify the number of DUTs into which one driver channel is branched to m. A driver channel can be branched only between the DUT with an odd number and the following DUT with an even number (i.e., between DUT1 and DUT2). For other combinations (i.e., between DUT1 and DUT3), a driver channel cannot be branched.

Available values

m = 1, 2, 4

Default

```
SEPARATE DR 1
```

◆ Example of a Correct Description

```
SEPARATE DR 2
DUT# = VS1, PD1-10, ...
DUT1 = PPS1, PD1-10A1, ...
DUT2 = PPS2, PD1-10A1, ...
DUT3 = PPS3, PD1-10B1, ...
DUT4 = PPS4, PD1-10B1, ...
```

◆ Example of an Incorrect Description

```
SEPARATE DR 2
DUT# = VS1, PD1-10, ...
DUT1 = PPS1, PD1-10A1, ...
DUT2 = PPS2, PD1-10B1, ...
DUT3 = PPS3, PD1-10A1*1, ...
DUT4 = PPS4, PD1-10B1, ...
```

**1 Cannot specify.*

(11) DUT# = VS_{n1}, PD_{n2}, PC_{n3}, P_{n4}

Specification of the channel number and the tester channel number of the VS (device power supply) written in test programs.

VS_{n1}

Specification of the channel number of the VS (device power supply). More than one VS channel numbers can be specified.

Available values

VS1 to VS8

Important

- Specify the same number of VSs as specified by PPS = b.
 - n_1 must be specified in ascending order.
 - When VS9 is specified by DEFINE VS or HVPPS, VS5 cannot be specified. When VS10 is specified by DEFINE VS or HVPPS, VS6 cannot be specified. When VS11 is specified by DEFINE VS or HVPPS, VS7 cannot be specified. When VS12 is specified by DEFINE VS or HVPPS, VS8 cannot be specified.
 - VS3 to VS8 can be used for T5593, T5592, T5591, T5591R, T5586, T5585, T5581P, T5571P, T5377, T5376, T5375, T5371, T5336 and T5311 systems.
-

Tip

VS can also be described as PPS.

PD n_2

Specification of driver channel number.

- ➔ For more information on available driver channel number, refer to "Tester channel allocation table".

PC n_3

Specification of comparator channel number.

- ➔ For more information on available comparator channel number, refer to "Tester channel allocation table".

P n_4

Specification of I/O channel number.

- ➔ For more information on available I/O channel number, refer to "Tester channel allocation table".

Important

The total number of channel numbers of PC n_3 and P n_4 must be the value of a specified by BIT = a or less.

(12) DUT k = PPS k_1 , PD k_2 xl, PC k_3 xl, P k_4 xl

Specification of PPS number and tester channel number for each DUT wired on actual performance board. Specify the PPS number and the tester channel number for each DUT corresponding to the channel number and the tester channel number of the VS specified in (11).

DUT k

Specification of DUT number.

Available values $k = 1$ to 256**PPSm₁**

Specification of PPS number.

Available values $m_1 = 1$ to 320 **Important**

- When the PARALLEL VS_n = m statement is written, specify m number of PPS numbers per VS_n.
- Multiple specification of the same PPS numbers is not possible.

 Tip*PPS can also be described as VS.***PDk₂xl**

Specification of driver channel number.

- ➔ For more information on available driver channel number for k₂, refer to "Tester channel allocation table".

 Important

When specifying a driver channel number, the number must start with either of the following numbers: 1, 11, 13, 17, 65, 75, 81, 129, 145, 193, 209, 257, 273, 321, 337, 385, 401, 449, or 465.

x

Specification of child name.

Available values

A, B, C, D, E, F, G, H

l

Specification of child number.

Available values $l = 1, 2, 3, 4, 5, 6, 7, 8$ **Default** $l = 1$

PCK₃x

Specification of comparator channel number.

- ➔ For more information on available comparator channel number for k₃, refer to "Tester channel allocation table".

x

Specification of child name.

Available values

A, B, C, D, E, F, G, H

l

Specification of child number.

Available values

l = 1, 3, 5, 7

Default

l = 1

Pk₄x

Specification of I/O channel number.

- ➔ For more information on available I/O channel number for k₄, refer to "Tester channel allocation table".

x

Specification of child name.

Available values

A, B, C, D, E, F, G, H

l

Specification of child number.

Available values

l = 1, 3, 5, 7

Default

l = 1

⚠ **Important**

Tester channels having the same child name (A, B, C, D, E, F, G, or H) must be allocated for one DUT. Therefore, the following specification is not possible. (A compilation error occurs.)

DUT1 = PD1-8A1, PD33B ← Cannot specify

◆ Consecutive Specification of Channel Numbers

With regard to the specification of PD_n, PC_n, and P_n in the preceding sections of (11) and (12), consecutive specification of channel numbers is possible by using a hyphen (-) as follows:

Figure 2-7 Consecutive Specification of Channel Numbers

PD_{n₁-m₁}A₁
PC_{n₂-m₂}A
P_{n₃-m₃}A

! Important

To perform consecutive specification, channel numbers must be specified in ascending order as follows: $n_1 < m_1$, $n_2 < m_2$, $n_3 < m_3$

2. 6 Addition of Revision Number to Object File at Compilation

2. 6. 1 %REMARK Statement

Function: Add revision information to the end of the object file at compilation. This statement can be written only once in a program. If this statement is written twice or more in a program, the second and subsequent %REMARK statements are ignored.

Syntax

```
%REMARK revision information
```

Argument: Revision information

Specification of character string to be added to the end of an object file as revision information. When the revision information exceeds 80 characters, the 81st and subsequent characters are ignored.

Available characters

Alphanumeric character, asterisk (*), underscore (_), point (.) and dollar sign (\$)

3. Allocation of Tester Channels and DC Measuring Units

This chapter describes the structure of the table (tester channel allocation table) showing driver channels and I/O channels to be allocated to the respective DUTs to test multiple DUTs simultaneously. The tester channel allocation table differs, depending on the type of test head. For details of tester channel allocation tables for respective test heads, refer to the subsequent chapters.

3. 1 Tester Channel Allocation Table

A tester channel allocation table shows the driver channels and I/O channels to be allocated to respective DUTs. Referring to the table, describe definitions of parallel test in the socket program. The following is an example of a tester channel allocation table.

Figure 3-1 Structure of Tester Channel Allocation Table

Tester channels allocated to one DUT

Table 6-1 PIN MODE = A 1 Child: 2

The number of DUTs to be tested simultaneously per test head

Driver Channel	I/O Channel	Number of DUTs for Parallel Test				
		1	2	4	8	16
PD1-32A1, PD31A1, PD32A1, PD65-96A1, PD95A1, PD96A1, PD157-160A1, PD221-224A1	P33-50A1 P97-114A1 P161-178A1 P225-242A1	1	2	4	8	16
PD1-32A3, PD31A3, PD32A3, PD65-96A3, PD95A3, PD96A3, PD157-160A3, PD221-224A3	P33-50A3 P97-114A3 P161-178A3 P225-242A3	----				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A)		A	A	A	A	A
		-	-	B	B	B
		-	-	-	C	C
		-	-	-	D	D
		-	-	-	-	E
		-	-	-	-	F
		-	-	-	-	G
		-	-	-	-	H

In this case, you cannot use the A3 side.

One DUT can be measured by using the channels of child A1.

Four DUTs can be measured by using the channels of child A1, A3, B1, and B3.

DUT pin number

Specify DUT numbers from DUT1 to DUTn (n = number of DUTs for parallel test) to the socket program. For example, when 64 DUTs are tested simultaneously, specify DUT1 to DUT64. DUT numbers can be freely specified regardless of tester channel numbers and child names.

! Important

When a high-precision calibration is used, some driver channels cannot be allocated to DUTs.

→ For more information on non-available driver channel number for DUT, refer to [T5500 Series Memory Test System Precision Calibration Program Reference Manual \(No. : 8311137\)](#) and [T5500 Series Memory Test System Precision Calibration2 Program Reference Manual \(No. : 8350393\)](#).

3. 2 DC Measuring Unit Number

The number of DC measuring units per test head can be 16DC, 32DC, or 64DC, depending on the type of the test head.

In case of 1600(576IO)CH, 800(288IO)CH, 1600(288IO)CH, 800(144IO)CH or 1344CH test head

There are two types: 16DC and 32DC. In case of 16DC, DC1 to DC16 are allocated to tester channels in the DC measuring unit allocation table in which DC17 to DC32 are written.

32DC	DC1 to DC16	DC17 to DC32
16DC	DC1 to DC16	DC1 to DC16

In case of 1664(576IO)CH/1664(640IO)CH test head

There are two types: 16DC and 32DC. In case of 16DC, DC1 to DC8 are allocated to tester channels in the DC measuring unit allocation table in which DC9 to DC16 are written, and DC9 to DC16 are allocated to tester channels in the DC measuring unit allocation table in which DC25 to DC32 are written.

32DC	DC1 to DC8	DC9 to DC16	DC17 to DC24	DC25 to DC32
16DC	DC1 to DC8	DC1 to DC8	DC9 to DC16	DC9 to DC16

In case of 2880CH test head

There are two types: 32DC and 64DC. In case of 32DC, DC1 to DC16 are allocated to tester channels in the DC measuring unit allocation table in which DC17 to DC32 are written, and DC33 to DC48 are allocated to tester channels in the DC measuring unit allocation table in which DC49 to DC64 are written.

64DC	DC1 to DC16	DC17 to DC32	DC33 to DC48	DC49 to DC64
32DC	DC1 to DC16	DC1 to DC16	DC33 to DC48	DC33 to DC48

◆ DC Multiplexer (DCMPX)

DC multiplexer (DCMPX), a hardware, is used to connect a DC test unit with tester channels of the test head. There are two types of DCMPX, Dn for driver channel side and Bn for I/O channel side. DCMPX operation is automatically controlled by the OS of the test system, based on the definition described in the socket program.

➔ For more information on DC measurement unit number connected for each DUT, refer to "DC measurement unit allocation table".

4. *1600(576IO)CH or 800(288IO)CH Test Head*

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 1600(576IO)CH or 800(288IO)CH test head.

4. 1 Test Head Configurations and Available Child Pins

1600(576IO)CH test head has two configurations, Full and Half. The child pins shown in the table below can be used for each test head.

Test head configuration		Child
1600(576IO)CH	Full	A, B, C, D, E, F, G, H
	Half	A, B, -, -, E, F, -, -
800(288IO)CH	-----	A, B, C, D, -, -, -, -

4. 2 DC Dedicated Channels and Option Driver Channels

DC dedicated channels and option driver channels in the table below are prepared.

Channel type	Channel
DC dedicated channel	31, 32, 95, 96
Option driver channel	21 to 30, 85 to 94

4. 3 Tester Channel Allocation Table (1)

Table 4-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1, PD65-96A1	P33-50A, P97-114A P161-178A, P225-242A	1	2	2	4	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
19 to 31	33-50, 97-109
32	33-48, 97-112
33 to 35	33-48, 97-112, 49, 50, 113
36	33-50, 97-114
37 to 63	33-50, 97-114, 161-178, 225-233
64	33-48, 97-112, 161-176, 225-240
65 to 71	33-48, 97-112, 161-176, 225-240, 49, 50, 113, 114, 177, 178, 241
72	33-50, 97-114, 161-178, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 4-8](#).

4. 4 Tester Channel Allocation Table (2)

Table 4-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1, PD65-96A1	P33-50A, P97-114A	2	4	4	8	8	16
PD1-32A2, PD65-96A2	P161-178A, P225-242A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
19 to 31	33-50, 97-109
	161-178, 225-237
32	33-48, 97-112
	161-176, 225-240
33 to 35	33-48, 97-112, 49, 50, 113
	161-176, 225-240, 177, 178, 241
36	33-50, 97-114
	161-178, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 4-8](#).

4. 5 Tester Channel Allocation Table (3)

Table 4-3 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1	P33-50A	4	8	8	16	16	32
PD65-96A1	P97-114A						
PD1-32A2	P161-178A						
PD65-96A2	P225-242A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

4. 5 Tester Channel Allocation Table (3)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
	97
	225
4	33-36
	161-164
	97-100
	225-228
8	33-40
	161-168
	97-104
	225-232
9	33-40, 49
	161-168, 177
	97-104, 113
	225-232, 241
16	33-48
	161-176
	97-112
	225-240
18	33-50
	161-178
	97-114
	225-242

➔ For more information on DC measurement unit allocation, refer to [Table 4-9](#).

4. 6 Tester Channel Allocation Table (4)

Table 4-4 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-10A1, PD21-25A1, PD31A1	P33-40A, P49A	8	16	16	32	32
PD11-20A1, PD26-30A1, PD32A1	P41-48A, P50A					
PD65-74A1, PD85-89A1, PD95A1	P97-104A, P113A					
PD75-84A1, PD90-94A1, PD96A1	P105-112A, P114A					
PD1-10A2, PD21-25A2, PD31A2	P161-168A, P177A					
PD11-20A2, PD26-30A2, PD32A2	P169-176A, P178A					
PD65-74A2, PD85-89A2, PD95A2	P225-232A, P241A					
PD75-84A2, PD90-94A2, PD96A2	P233-240A, P242A					
Child names to be used (When child B to F are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A
		-	B	-	B	B
		-	-	-	C	-
		-	-	-	D	-
		-	-	E	-	E
		-	-	-	-	F
		-	-	-	-	-
		-	-	-	-	-

4. 6 Tester Channel Allocation Table (4)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
	41
	169
	97
	225
	105
	233
4	33-36
	161-164
	41-44
	169-172
	97-100
	225-228
	105-108
	233-236
8	33-40
	161-168
	41-48
	169-176
	97-104
	225-232
	105-112
	233-240
9	33-40, 49
	161-168, 177
	41-48, 50
	169-176, 178
	97-104, 113
	225-232, 241
	105-112, 114
	233-240, 242

➔ For more information on DC measurement unit allocation, refer to [Table 4-9](#).

4. 7 Tester Channel Allocation Table (5)

Important

This allocation table can be used only for 1600(576I/O)CH test head.

Table 4-5 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test											
PD1-32A1, PD65-96A1	P33-50A, P161-178A	1	2	2	2	4	4	4	4	8	8	8	16
PD1-32A2, PD65-96A2	P97-114A, P225-242A	---		---	---		---		---			---	
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A	A	A	A	A	A	A
		-	-	B	-	B	B	B	B	B	B	B	B
		-	-	-	-	-	C	-	-	C	-	C	C
		-	-	-	-	-	D	-	-	D	-	D	D
		-	-	-	E	-	-	-	E	-	E	E	E
		-	-	-	-	-	-	-	F	-	F	F	F
		-	-	-	-	-	-	-	-	-	-	G	G
		-	-	-	-	-	-	-	-	-	-	H	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
19 to 31	33-50, 161-173
	97-114, 225-237
32	33-48, 161-176
	97-112, 225-240
33 to 35	33-48, 161-176, 49, 50, 177
	97-112, 225-240, 113, 114, 241
36	33-50, 161-178
	97-114, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 4-8](#).

4. 8 Tester Channel Allocation Table (6)

Table 4-6 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1	P33-50A, P97-114A	2	4	4	8	8	16
PD65-96A2	P161-178A, P225-242A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
19 to 31	33-50, 97-109
	161-178, 225-237
32	33-48, 97-112
	161-176, 225-240
33 to 35	33-48, 97-112, 49, 50, 113
	161-176, 225-240, 177, 178, 241
36	33-50, 97-114
	161-178, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 4-8](#).

4. 9 Tester Channel Allocation Table (7)

Important

This allocation table can be used only for 1600(576IO)CH test head.

Table 4-7 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-32A1	P33-40A, P49A	64	-----	-----
PD1-32A1	P41-48A, P50A			
PD65-96A1	P97-104A, P113A			
PD65-96A1	P105-112A, P114A			
PD1-32A2	P161-168A, P177A			
PD1-32A2	P169-176A, P178A			
PD65-96A2	P225-232A, P241A			
PD65-96A2	P233-240A, P242A			
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----

4. 9 Tester Channel Allocation Table (7)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
	41
	169
	97
	225
	105
	233
4	33-36
	161-164
	41-44
	169-172
	97-100
	225-228
	105-108
	233-236
8	33-40
	161-168
	41-48
	169-176
	97-104
	225-232
	105-112
	233-240
9	33-40, 49
	161-168, 177
	41-48, 50
	169-176, 178
	97-104, 113
	225-232, 241
	105-112, 114
	233-240, 242

➔ For more information on DC measurement unit allocation, refer to [Table 4-9](#).

4.10 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 4-8 DC Measuring Unit Allocation Table (1)

DCMPX = D1, B18 16DC MODE						
Child	Tester channel					
	1-32	65-96	33-50	97-114	161-178	225-242
A1	DC1		DC1		DC2	
A2	DC2		-----			
A3	-----					
A4	-----					
B1	DC3		DC3		DC4	
B2	DC4		-----			
B3	-----					
B4	-----					
C1	DC5		DC5		DC6	
C2	DC6		-----			
C3	-----					
C4	-----					
D1	DC7		DC7		DC8	
D2	DC8		-----			
D3	-----					
D4	-----					
E1	DC9		DC9		DC10	
E2	DC10		-----			
E3	-----					
E4	-----					
F1	DC11		DC11		DC12	
F2	DC12		-----			
F3	-----					
F4	-----					
G1	DC13		DC13		DC14	
G2	DC14		-----			
G3	-----					
G4	-----					
H1	DC15		DC15		DC16	
H2	DC16		-----			
H3	-----					
H4	-----					

4. 10 DC Measuring Unit Allocation Table

Table 4-9 DC Measuring Unit Allocation Table (2)

DCMPX = D1, B18 16DC/32DC MODE						
Child	Tester channel					
	1-32	65-96	33-50	97-114	161-178	225-242
A1	DC1	DC17	DC1	DC17	DC2	DC18
A2	DC2	DC18	-----			
A3	-----					
A4	-----					
B1	DC3	DC19	DC3	DC19	DC4	DC20
B2	DC4	DC20	-----			
B3	-----					
B4	-----					
C1	DC5	DC21	DC5	DC21	DC6	DC22
C2	DC6	DC22	-----			
C3	-----					
C4	-----					
D1	DC7	DC23	DC7	DC23	DC8	DC24
D2	DC8	DC24	-----			
D3	-----					
D4	-----					
E1	DC9	DC25	DC9	DC25	DC10	DC26
E2	DC10	DC26	-----			
E3	-----					
E4	-----					
F1	DC11	DC27	DC11	DC27	DC12	DC28
F2	DC12	DC28	-----			
F3	-----					
F4	-----					
G1	DC13	DC29	DC13	DC29	DC14	DC30
G2	DC14	DC30	-----			
G3	-----					
G4	-----					
H1	DC15	DC31	DC15	DC31	DC16	DC32
H2	DC16	DC32	-----			
H3	-----					
H4	-----					

5. *1600(288IO)CH or 800(144IO)CH Test Head*

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 1600(288IO)CH or 800(144IO)CH test head.

5. 1 Test Head Configurations and Available Child Pins

1600(288IO)CH test head has two configurations, Full and Half. The child pins shown in the table below can be used for each test head.

Test head configuration		Child
1600(288IO)CH	Full	A, B, C, D, E, F, G, H
	Half	A, B, -, -, E, F, -, -
800(144IO)CH	-----	A, B, C, D, -, -, -, -

5. 2 DC Dedicated Channels and Option Driver Channels

DC dedicated channels and option driver channels in the table below are prepared.

Channel type	Channel
DC dedicated channel	31, 32, 95, 96
Option driver channel	21 to 30, 85 to 94

5. 3 Tester Channel Allocation Table (1)

Table 5-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1, PD65-96A1	P33-50A, P97-114A	1	2	2	4	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
19 to 31	33-50, 97-109
32	33-48, 97-112
33 to 35	33-48, 97-112, 49, 50, 113
36	33-50, 97-114

➔ For more information on DC measurement unit allocation, refer to [Table 5-5](#).

5. 4 Tester Channel Allocation Table (2)

Table 5-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1, PD65-96A1	P33-50A	2	4	4	8	8	16
PD1-32A2, PD65-96A2	P97-114A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114

➔ For more information on DC measurement unit allocation, refer to [Table 5-6](#).

5. 5 Tester Channel Allocation Table (3)

Table 5-3 PIN MODE = B 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1	P33-40A, P49A	4	8	8	16	16	32
PD65-96A1	P41-48A, P50A						
PD1-32A2	P97-104A, P113A						
PD65-96A2	P105-112A, P114A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	41
	105
4	33-36
	97-100
	41-44
	105-108
8	33-40
	97-104
	41-48
	105-112
9	33-40, 49
	97-104, 113
	41-48, 50
	105-112, 114

➔ For more information on DC measurement unit allocation, refer to [Table 5-7](#).

5. 6 Tester Channel Allocation Table (4)

Table 5-4 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test					
PD1-32A1	P33-50A	2	4	4	8	8	16
PD65-96A2	P97-114A						
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A
		-	B	-	B	B	B
		-	-	-	C	-	C
		-	-	-	D	-	D
		-	-	E	-	E	E
		-	-	-	-	F	F
		-	-	-	-	-	G
		-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114

➔ For more information on DC measurement unit allocation, refer to [Table 5-6](#).

5. 7 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 5-5 DC Measuring Unit Allocation Table (1)

DCMPX = D1, B18 16DC MODE						
Child	Tester channel					
	1-32	65-96	33-40, 49	41-48, 50	97-104, 113	105-112, 114
A1	DC1					
A2	-----					
A3	-----					
A4	-----					
B1	DC3					
B2	-----					
B3	-----					
B4	-----					
C1	DC5					
C2	-----					
C3	-----					
C4	-----					
D1	DC7					
D2	-----					
D3	-----					
D4	-----					
E1	DC9					
E2	-----					
E3	-----					
E4	-----					
F1	DC11					
F2	-----					
F3	-----					
F4	-----					
G1	DC13					
G2	-----					
G3	-----					
G4	-----					
H1	DC15					
H2	-----					
H3	-----					
H4	-----					

Table 5-6 DC Measuring Unit Allocation Table (2)

DCMPX = D1, B9 16DC MODE						
Child	Tester channel					
	1-32	65-96	33-40, 49	41-48, 50	97-104, 113	105-112, 114
A1	DC1		DC1		DC2	
A2	DC2		-----			
A3	-----					
A4	-----					
B1	DC3		DC3		DC4	
B2	DC4		-----			
B3	-----					
B4	-----					
C1	DC5		DC5		DC6	
C2	DC6		-----			
C3	-----					
C4	-----					
D1	DC7		DC7		DC8	
D2	DC8		-----			
D3	-----					
D4	-----					
E1	DC9		DC9		DC10	
E2	DC10		-----			
E3	-----					
E4	-----					
F1	DC11		DC11		DC12	
F2	DC12		-----			
F3	-----					
F4	-----					
G1	DC13		DC13		DC14	
G2	DC14		-----			
G3	-----					
G4	-----					
H1	DC15		DC15		DC16	
H2	DC16		-----			
H3	-----					
H4	-----					

Table 5-7 DC Measuring Unit Allocation Table (3)

DCMPX = D1, B9 16DC/32DC MODE						
Child	Tester Channel					
	1-32	65-96	33-40, 49	41-48, 50	97-104, 113	105-112, 114
A1	DC1	DC17	DC1	DC17	DC2	DC18
A2	DC2	DC18	-----			
A3	-----					
A4	-----					
B1	DC3	DC19	DC3	DC19	DC4	DC20
B2	DC4	DC20	-----			
B3	-----					
B4	-----					
C1	DC5	DC21	DC5	DC21	DC6	DC22
C2	DC6	DC22	-----			
C3	-----					
C4	-----					
D1	DC7	DC23	DC7	DC23	DC8	DC24
D2	DC8	DC24	-----			
D3	-----					
D4	-----					
E1	DC9	DC25	DC9	DC25	DC10	DC26
E2	DC10	DC26	-----			
E3	-----					
E4	-----					
F1	DC11	DC27	DC11	DC27	DC12	DC28
F2	DC12	DC28	-----			
F3	-----					
F4	-----					
G1	DC13	DC29	DC13	DC29	DC14	DC30
G2	DC14	DC30	-----			
G3	-----					
G4	-----					
H1	DC15	DC31	DC15	DC31	DC16	DC32
H2	DC16	DC32	-----			
H3	-----					
H4	-----					

6. *1664(576IO)CH or 1664(640IO)CH Test Head*

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 1664(576IO)CH or 1664(640IO)CH test head.

6. 1 Test Head Configurations and Available Child Pins

1664(576IO)CH test head has two configurations, Full and Half, and 1664(640IO)CH test head has three configurations, Full, Half, and Quarter. The child pins shown in the table below can be used for each configuration.

Test head configuration		Child	
1664(576IO)CH	Full	A1, A2, A3, A4	B1, B2, B3, B4
		C1, C2, C3, C4	D1, D2, D3, D4
		E1, E2, E3, E4	F1, F2, F3, F4
		G1, G2, G3, G4	H1, H2, H3, H4
	Half	A1, A2, A3, A4	B1, B2, B3, B4
		C1, C2, C3, C4	D1, D2, D3, D4
		-----	-----
		-----	-----
1664(640IO)CH	Full	A1, A2, A3, A4	B1, B2, B3, B4
		C1, C2, C3, C4	D1, D2, D3, D4
		E1, E2, E3, E4	F1, F2, F3, F4
		G1, G2, G3, G4	H1, H2, H3, H4
	Half	A1, A2, A3, A4	B1, B2, B3, B4
		C1, C2, C3, C4	D1, D2, D3, D4
		-----	-----
		-----	-----
	Quarter	A1, A2, --, --	B1, B2, --, --
		C1, C2, --, --	D1, D2, --, --
		-----	-----
		-----	-----

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

51, 52, 115, 116

6. 2 DC Dedicated Channels and Driver Dedicated Channels

DC dedicated channels and driver dedicated channels in the table below are provided.

Channel type	Channel
DC dedicated channel	31, 32
Driver dedicated channel ^{*1}	51, 52, 115, 116

**1 Only in case of 1664(576IO)CH test head*

6. 3 Tester Channel Allocation Table (1)

Table 6-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test						
PD1-32A1	P33-52A1, P97-116A1	1	2	2	2	4	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A	A
		-	B	-	-	B	-	B
		-	-	C	-	C	C	C
		-	-	-	-	D	-	D
		-	-	-	E	-	E	E
		-	-	-	-	-	-	F
		-	-	-	-	-	G	G
		-	-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
10 ^{*1}	33-40, 49, 51
16	33-48
18	33-50
19	33-50, 97
20	33-50, 97-98
20	33-52 ^{*1}
21 to 31	33-50, 97-109
32	33-48, 97-112
33 to 35	33-50, 97-113
36	33-50, 97-114
40 ^{*1}	33-52, 97-116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-9](#).

6. 4 Tester Channel Allocation Table (2)

Table 6-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-32A1	P33-52A1, P97-116A1	2	4	8	8	16
PD1-32A3	P33-52A3, P97-116A3					
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A
		-	B	B	-	B
		-	-	C	C	C
		-	-	D	-	D
		-	-	-	E	E
		-	-	-	-	F
		-	-	-	G	G
		-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
10 ^{*1}	33-40, 49, 51
16	33-48
18	33-50
19	33-50, 97
20	33-50, 97-98
20	33-52 ^{*1}
21 to 31	33-50, 97-109
32	33-48, 97-112
33 to 35	33-50, 97-113
36	33-50, 97-114
40 ^{*1}	33-52, 97-116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-10](#).

6. 5 Tester Channel Allocation Table (3)

Table 6-3 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test						
PD1-32A1	P33-52A1	2	4	4	4	8	8	16
PD1-32A2	P97-116A1							
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A	A
		-	B	-	-	B	-	B
		-	-	C	-	C	C	C
		-	-	-	-	D	-	D
		-	-	-	E	-	E	E
		-	-	-	-	-	-	F
		-	-	-	-	-	G	G
		-	-	-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
10 ^{*1}	33-40, 49, 51
	97-104, 113, 115
16	33-48
	97-112
18	33-50
	97-114
20 ^{*1}	33-52
	97-116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6. 6 Tester Channel Allocation Table (4)

Table 6-4 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-32A1	P33-52A1	4	8	16	16	32
PD1-32A2	P97-116A1					
PD1-32A3	P33-52A3					
PD1-32A4	P97-116A3					
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A
		-	B	B	-	B
		-	-	C	C	C
		-	-	D	-	D
		-	-	-	E	E
		-	-	-	-	F
		-	-	-	G	G
		-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
	33-40, 49A3
	97-104, 113A3
10 ^{*1}	33-40, 49, 51
	97-104, 113, 115
16	33-48
	97-112
18	33-50
	97-114
20 ^{*1}	33-52
	97-116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6. 7 Tester Channel Allocation Table (5)

Table 6-5 PIN MODE = B 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-12A1, PD25-27A1, PD31A1	P33-40A1, P49A1, P51A1	8	8	16	16	32
PD13-24A1, PD28-30A1, PD32A1	P41-48A1, P50A1, P52A1					
PD1-12A2, PD25-27A2, PD31A2	P97-104A1, P113A1, P115A1					
PD13-24A2, PD28-30A2, PD32A2	P105-112A1, P114A1, P116A1					
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A
		-	-	B	-	B
		C	-	C	C	C
		-	-	D	-	D
		-	E	-	E	E
		-	-	-	-	F
		-	-	-	G	G
		-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10 ^{*1}	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6. 8 Tester Channel Allocation Table (6)

Table 6-6 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-12A1, PD25-27A1, PD31A1	P33-40A1, P49A1, P51A1	8	16	32	32	64 ^{*1}
PD13-24A1, PD28-30A1, PD32A1	P41-48A1, P50A1, P52A1					
PD1-12A2, PD25-27A2, PD31A2	P97-104A1, P113A1, P115A1					
PD13-24A2, PD28-30A2, PD32A2	P105-112A1, P114A1, P116A1					
PD1-12A3, PD25-27A3, PD31A3	P33-40A3, P49A3, P51A3					
PD13-24A3, PD28-30A3, PD32A3	P41-48A3, P50A3, P52A3					
PD1-12A4, PD25-27A4, PD31A4	P97-104A3, P113A3, P115A3					
PD13-24A4, PD28-30A4, PD32A4	P105-112A3, P114A3, P116A3					
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A
		-	B	B	-	B
		-	-	C	C	C
		-	-	D	-	D
		-	-	-	E	E
		-	-	-	-	F
		-	-	-	G	G
		-	-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10 ^{*1}	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

^{*1} Available only for 1664 (640IO) CH test head.

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6. 9 Tester Channel Allocation Table (7)

! Important

This allocation table can be used only for 1664(640IO)CH test head.

Table 6-7 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-32A1	P33-40A1, P49A1, P51A1	64	-----	-----
PD1-32A1	P41-48A1, P50A1, P52A1			
PD1-32A2	P97-104A1, P113A1, P115A1			
PD1-32A2	P105-112A1, P114A1, P116A1			
PD1-32A3	P33-40A3, P49A3, P51A3			
PD1-32A3	P41-48A3, P50A3, P52A3			
PD1-32A4	P97-104A3, P113A3, P115A3			
PD1-32A4	P105-112A3, P114A3, P116A3			
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6.10 Tester Channel Allocation Table (8)

Important

This allocation table can be used only for 1664(640IO)CH test head and 32DC.

Table 6-8 PIN MODE = C 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-8A1, PD49A1	P33-36A1	128	-----	-----
PD9-12A1, PD25-27A1, PD31A1, PD51A1	P37-40A1			
PD13-20A1, PD50A1	P41-44A1			
PD21-24A1, PD28-30A1, PD32A1, PD52A1	P45-48A1			
PD1-8A2, PD113A1	P97-100A1			
PD9-12A2, PD25-27A2, PD31A2, PD115A1	P101-104A1			
PD13-20A2, PD114A1	P105-108A1			
PD21-24A2, PD28-30A2, PD32A2, PD116A1	P109-112A1			
PD1-8A3, PD49A3	P33-36A3			
PD9-12A3, PD25-27A3, PD31A3, PD51A3	P37-40A3			
PD13-20A3, PD50A3	P41-44A3			
PD21-24A3, PD28-30A3, PD32A3, PD52A3	P45-48A3			
PD1-8A4, PD113A3	P97-100A3			
PD9-12A4, PD25-27A4, PD31A4, PD115A3	P101-104A3			
PD13-20A4, PD114A3	P105-108A3			
PD21-24A4, PD28-30A4, PD32A4, PD116A3	P109-112A3			
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	37
	41
	45
	97
	101
	105
	109
4	33-36
	37-40
	41-44
	45-48
	97-100
	101-104
	105-108
	109-112

➔ For more information on DC measurement unit allocation, refer to [Table 6-11](#).

6.11 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 6-9 DC Measuring Unit Allocation Table (1)

DCMPX = D1, B36 32DC MODE										
Child	Tester channel									
	1-12	13-24	25-27	28-30	31	32	33-40, 49, 51	41-48, 50, 52	97-104, 113, 115	105-112, 114, 116
A1	DC1									
A2	-----									
A3	-----									
A4	-----									
B1	DC3									
B2	-----									
B3	-----									
B4	-----									
C1	DC5									
C2	-----									
C3	-----									
C4	-----									
D1	DC7									
D2	-----									
D3	-----									
D4	-----									
E1	DC17									
E2	-----									
E3	-----									
E4	-----									
F1	DC19									
F2	-----									
F3	-----									
F4	-----									
G1	DC21									
G2	-----									
G3	-----									
G4	-----									
H1	DC23									
H2	-----									
H3	-----									
H4	-----									

Table 6-10 DC Measuring Unit Allocation Table (2)

DCMPX = D1, B36 32DC MODE										
Child	Tester channel									
	1-12	13-24	25-27	28-30	31	32	33-40, 49, 51	41-48, 50, 52	97-104, 113, 115	105-112, 114, 116
A1	DC1									
A2	-----									
A3	DC2									
A4	-----									
B1	DC3									
B2	-----									
B3	DC4									
B4	-----									
C1	DC5									
C2	-----									
C3	DC6									
C4	-----									
D1	DC7									
D2	-----									
D3	DC8									
D4	-----									
E1	DC9									
E2	-----									
E3	DC10									
E4	-----									
F1	DC11									
F2	-----									
F3	DC12									
F4	-----									
G1	DC13									
G2	-----									
G3	DC14									
G4	-----									
H1	DC15									
H2	-----									
H3	DC16									
H4	-----									

Table 6-11 DC Measuring Unit Allocation Table (3)

DCMPX = D1, B18 32DC MODE										
Child	Tester channel									
	1-12	13-24	25-27	28-30	31	32	33-40, 49, 51	41-48, 50, 52	97-104, 113, 115	105-112, 114, 116
A1	DC1						DC1		DC9	
A2	DC9						-----			
A3	DC2						DC2		DC10	
A4	DC10						-----			
B1	DC3						DC3		DC11	
B2	DC11						-----			
B3	DC4						DC4		DC12	
B4	DC12						-----			
C1	DC5						DC5		DC13	
C2	DC13						-----			
C3	DC6						DC6		DC14	
C4	DC14						-----			
D1	DC7						DC7		DC15	
D2	DC15						-----			
D3	DC8						DC8		DC16	
D4	DC16						-----			
E1	DC17						DC17		DC25	
E2	DC25						-----			
E3	DC18						DC18		DC26	
E4	DC26						-----			
F1	DC19						DC19		DC27	
F2	DC27						-----			
F3	DC20						DC20		DC28	
F4	DC28						-----			
G1	DC21						DC21		DC29	
G2	DC29						-----			
G3	DC22						DC22		DC30	
G4	DC30						-----			
H1	DC23						DC23		DC31	
H2	DC31						-----			
H3	DC24						DC24		DC32	
H4	DC32						-----			

7. 352CH Test Head

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 352CH test head.

7. 1 Test Head Configurations and Available Tester Channels

Test head configuration				Driver channel	I/O channel
352CH	Full	Half	Quarter	PD1-11A1 PD17-27A1 PD65-75A1 PD81-91A1	P33-43A P49-59A P97-107A P113-123A
				PD129-139A1 PD145-155A1 PD193-203A1 PD209-219A1	P161-171A P177-187A P225-235A P241-251A
				PD257-267A1 PD273-283A1 PD321-331A1 PD337-347A1 PD385-395A1 PD401-411A1 PD449-459A1 PD465-475A1	P289-299A P305-315A P353-363A P369-379A P417-427A P433-443A P481-491A P497-507A

Important

This test head has only child A.

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

42, 43, 58, 59, 106, 107, 122, 123, 170, 171, 186, 187, 234, 235, 250, 251, 298, 299,
314, 315, 362, 363, 378, 379, 426, 427, 442, 443, 490, 491, 506, 507

7. 2 Tester Channel Allocation Table of T5591

7. 2. 1 Tester Channel Allocation Table (1)

Table 7-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-11A1 PD17-27A1 PD65-75A1 PD81-91A1 PD129-139A1 PD145-155A1 PD193-203A1 PD209-219A1 PD257-267A1 PD273-283A1 PD321-331A1 PD337-347A1 PD385-395A1 PD401-411A1 PD449-459A1 PD465-475A1	P33-43A P49-59A P97-107A P113-123A P161-171A P177-187A P225-235A P241-251A P289-299A P305-315A P353-363A P369-379A P417-427A P433-443A P481-491A P497-507A	1	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
8	33-40
9	33-41
11	33-43
16	33-40, 49-56
18	33-41, 49-57
22	33-43, 49-59
32	33-40, 49-56, 97-104, 113-120
36	33-41, 49-57, 97-105, 113-121
44	33-43, 49-59, 97-107, 113-123
64	33-40, 49-56, 97-104, 113-120, 161-168, 177-184, 225-232, 241-248
72	33-41, 49-57, 97-105, 113-121, 161-169, 177-185, 225-233, 241-249
88	33-43, 49-59, 97-107, 113-123, 161-171, 177-187, 225-235, 241-251
128	33-40, 49-56, 97-104, 113-120, 161-168, 177-184, 225-232, 241-248 289-296, 305-312, 353-360, 369-376, 417-424, 433-440, 481-488, 497-504
144	33-41, 49-57, 97-105, 113-121, 161-169, 177-185, 225-233, 241-249 289-297, 305-313, 353-361, 369-377, 417-425, 433-441, 481-489, 497-505
176	33-43, 49-59, 97-107, 113-123, 161-171, 177-187, 225-235, 241-251 289-299, 305-315, 353-363, 369-379, 417-427, 433-443, 481-491, 497-507

7. 2. 2 Tester Channel Allocation Table (2)

Table 7-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-11A1 PD17-27A1 PD65-75A1 PD81-91A1 PD129-139A1 PD145-155A1 PD193-203A1 PD209-219A1	P33-43A P49-59A P97-107A P113-123A P161-171A P177-187A P225-235A P241-251A	2	-----	-----
PD257-267A1 PD273-283A1 PD321-331A1 PD337-347A1 PD385-395A1 PD401-411A1 PD449-459A1 PD465-475A1	P289-299A P305-315A P353-363A P369-379A P417-427A P433-443A P481-491A P497-507A			

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	289-292
8	33-40
	289-296
9	33-41
	289-297
11	33-43
	289-299
16	33-40, 49-56
	289-296, 305-312
18	33-41, 49-57
	289-297, 305-313
22	33-43, 49-59
	289-299, 305-315
32	33-40, 49-56, 97-104, 113-120
	289-296, 305-312, 353-360, 369-376
36	33-41, 49-57, 97-105, 113-121
	289-297, 305-313, 353-361, 369-377
44	33-43, 49-59, 97-107, 113-123
	289-299, 305-315, 353-363, 369-379
64	33-40, 49-56, 97-104, 113-120, 161-168, 177-184, 225-232, 241-248
	289-296, 305-312, 353-360, 369-376, 417-424, 433-440, 481-488, 497-504
72	33-41, 49-57, 97-105, 113-121, 161-169, 177-185, 225-233, 241-249
	289-297, 305-313, 353-361, 369-377, 417-425, 433-441, 481-489, 497-505
88	33-43, 49-59, 97-107, 113-123, 161-171, 177-187, 225-235, 241-251
	289-299, 305-315, 353-363, 369-379, 417-427, 433-443, 481-491, 497-507

7. 2. 3 Tester Channel Allocation Table (3)

Table 7-3 PIN MODE = A 1 Child: 2/4

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-11A1 PD17-27A1 PD65-75A1 PD81-91A1	P33-43A P49-59A P97-107A P113-123A	2	4	-----
PD129-139A1 PD145-155A1 PD193-203A1 PD209-219A1	P161-171A P177-187A P225-235A P241-251A			
PD257-267A1 PD273-283A1 PD321-331A1 PD337-347A1	P289-299A P305-315A P353-363A P369-379A	-----		
PD385-395A1 PD401-411A1 PD449-459A1 PD465-475A1	P417-427A P433-443A P481-491A P497-507A			

7. 2 Tester Channel Allocation Table of T5591

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	161-164
	289-292
	417-420
8	33-40
	161-168
	289-296
	417-424
9	33-41
	161-169
	289-297
	417-425
11	33-43
	161-171
	289-299
	417-427
16	33-40, 49-56
	161-168, 177-184
	289-296, 305-312
	417-424, 433-440
18	33-41, 49-57
	161-169, 177-185
	289-297, 305-313
	417-425, 433-441
22	33-43, 49-59
	161-171, 177-187
	289-299, 305-315
	417-427, 433-443
32	33-40, 49-56, 97-104, 113-120
	161-168, 177-184, 225-232, 241-248
	289-296, 305-312, 353-360, 369-376
	417-424, 433-440, 481-488, 497-504
36	33-41, 49-57, 97-105, 113-121
	161-169, 177-185, 225-233, 241-249
	289-297, 305-313, 353-361, 369-377
	417-425, 433-441, 481-489, 497-505
44	33-43, 49-59, 97-107, 113-123
	161-171, 177-187, 225-235, 241-251
	289-299, 305-315, 353-363, 369-379
	417-427, 433-443, 481-491, 497-507

7. 2. 4 Tester Channel Allocation Table (4)

Table 7-4 PIN MODE = A 1 Child: 2/4/8

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-11A1 PD17-27A1	P33-43A P49-59A	2	4	8
PD65-75A1 PD81-91A1	P97-107A P113-123A			
PD129-139A1 PD145-155A1	P161-171A P177-187A	-----		
PD193-203A1 PD209-219A1	P225-235A P241-251A			
PD257-267A1 PD273-283A1	P289-299A P305-315A	-----	-----	
PD321-331A1 PD337-347A1	P353-363A P369-379A			
PD385-395A1 PD401-411A1	P417-427A P433-443A	-----	-----	
PD449-459A1 PD465-475A1	P481-491A P497-507A			

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	97-100
	161-164
	225-228
	289-292
	353-356
	417-420
	481-484
8	33-40
	97-104
	161-168
	225-232
	289-296
	353-360
	417-424
	481-488
9	33-41
	97-105
	161-169
	225-233
	289-297
	353-361
	417-425
	481-489
11	33-43
	97-107
	161-171
	225-235
	289-299
	353-363
	417-427
	481-491
16	33-40, 49-56
	97-104, 113-120
	161-168, 177-184
	225-232, 241-248
	289-296, 305-312
	353-360, 369-376
	417-424, 433-440
	481-488, 497-504

7. 2 Tester Channel Allocation Table of T5591

Bit (BIT=a)	Available I/O channel
18	33-41, 49-57
	97-105, 113-121
	161-169, 177-185
	225-233, 241-249
	289-297, 305-313
	353-361, 369-377
	417-425, 433-441
	481-489, 497-505
22	33-43, 49-59
	97-107, 113-123
	161-171, 177-187
	225-235, 241-251
	289-299, 305-315
	353-363, 369-379
	417-427, 433-443
	481-491, 497-507

7. 2. 5 Tester Channel Allocation Table (5)

Table 7-5 PIN MODE = B 1 Child: 2/4/8/16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-11A1	P33-43A	2	4	8	16
PD17-27A1	P49-59A				
PD65-75A1	P97-107A	-----	-----	-----	
PD81-91A1	P113-123A	-----			
PD129-139A1	P161-171A	-----	-----	-----	
PD145-155A1	P177-187A	-----			
PD193-203A1	P225-235A	-----	-----	-----	
PD209-219A1	P241-251A	-----			
PD257-267A1	P289-299A	-----	-----	-----	
PD273-283A1	P305-315A	-----			
PD321-331A1	P353-363A	-----	-----	-----	
PD337-347A1	P369-379A	-----			
PD385-395A1	P417-427A	-----	-----	-----	
PD401-411A1	P433-443A	-----			
PD449-459A1	P481-491A	-----	-----	-----	
PD465-475A1	P497-507A	-----			

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	49-52
	97-100
	113-116
	161-164
	177-180
	225-228
	241-244
	289-292
	305-308
	353-356
	369-372
	417-420
	433-436
	481-484
	497-500
8	33-40
	49-56
	97-104
	113-120
	161-168
	177-184
	225-232
	241-248
	289-296
	305-312
	353-360
	369-376
	417-424
	433-440
	481-488
	497-504

7. 2 Tester Channel Allocation Table of T5591

Bit (BIT=a)	Available I/O channel
9	33-41
	49-57
	97-105
	113-121
	161-169
	177-185
	225-233
	241-249
	289-297
	305-313
	353-361
	369-377
	417-425
	433-441
	481-489
	497-505
11	33-43
	49-59
	97-107
	113-123
	161-171
	177-187
	225-235
	241-251
	289-299
	305-315
	353-363
	369-379
	417-427
	433-443
	481-491
	497-507

7. 3 Tester Channel Allocation Table of T5591R

7. 3. 1 Tester Channel Allocation Table (1)

Table 7-6 PIN MODE = A 1 Child: 1/2/4/8

Driver channel	I/O channel	Number of DUTs for parallel test			
		1	2	4	8
PD1-11A1 PD17-27A1	P33-43A P49-59A	1	2	4	8
PD65-75A1 PD81-91A1	P97-107A P113-123A	-----	-----	-----	-----
PD129-139A1 PD145-155A1	P161-171A P177-187A	-----	-----	-----	-----
PD193-203A1 PD209-219A1	P225-235A P241-251A	-----	-----	-----	-----
PD257-267A1 PD273-283A1	P289-299A P305-315A	-----	-----	-----	-----
PD321-331A1 PD337-347A1	P353-363A P369-379A	-----	-----	-----	-----
PD385-395A1 PD401-411A1	P417-427A P433-443A	-----	-----	-----	-----
PD449-459A1 PD465-475A1	P481-491A P497-507A	-----	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	97-100
	161-164
	225-228
	289-292
	353-356
	417-420
	481-484
8	33-40
	97-104
	161-168
	225-232
	289-296
	353-360
	417-424
	481-488
9	33-41
	97-105
	161-169
	225-233
	289-297
	353-361
	417-425
	481-489
11	33-43
	97-107
	161-171
	225-235
	289-299
	353-363
	417-427
	481-491
16	33-40, 49-56
	97-104, 113-120
	161-168, 177-184
	225-232, 241-248
	289-296, 305-312
	353-360, 369-376
	417-424, 433-440
	481-488, 497-504

7. 3 Tester Channel Allocation Table of T5591R

Bit (BIT=a)	Available I/O channel
18	33-41, 49-57
	97-105, 113-121
	161-169, 177-185
	225-233, 241-249
	289-297, 305-313
	353-361, 369-377
	417-425, 433-441
	481-489, 497-505
22	33-43, 49-59
	97-107, 113-123
	161-171, 177-187
	225-235, 241-251
	289-299, 305-315
	353-363, 369-379
	417-427, 433-443
	481-491, 497-507

7. 3. 2 Tester Channel Allocation Table (2)

Table 7-7 PIN MODE = B 1 Child: 2/4/8/16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-11A1	P33-43A	2	4	8	16
PD17-27A1	P49-59A				
PD65-75A1	P97-107A	-----	-----	-----	
PD81-91A1	P113-123A	-----			
PD129-139A1	P161-171A	-----	-----	-----	
PD145-155A1	P177-187A	-----			
PD193-203A1	P225-235A	-----	-----	-----	
PD209-219A1	P241-251A	-----			
PD257-267A1	P289-299A	-----	-----	-----	
PD273-283A1	P305-315A	-----			
PD321-331A1	P353-363A	-----	-----	-----	
PD337-347A1	P369-379A	-----			
PD385-395A1	P417-427A	-----	-----	-----	
PD401-411A1	P433-443A	-----			
PD449-459A1	P481-491A	-----	-----	-----	
PD465-475A1	P497-507A	-----			

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
4	33-36
	49-52
	97-100
	113-116
	161-164
	177-180
	225-228
	241-244
	289-292
	305-308
	353-356
	369-372
	417-420
	433-436
	481-484
	497-500
8	33-40
	49-56
	97-104
	113-120
	161-168
	177-184
	225-232
	241-248
	289-296
	305-312
	353-360
	369-376
	417-424
	433-440
	481-488
	497-504

7. 3 Tester Channel Allocation Table of T5591R

Bit (BIT=a)	Available I/O channel
9	33-41
	49-57
	97-105
	113-121
	161-169
	177-185
	225-233
	241-249
	289-297
	305-313
	353-361
	369-377
	417-425
	433-441
	481-489
	497-505
11	33-43
	49-59
	97-107
	113-123
	161-171
	177-187
	225-235
	241-251
	289-299
	305-315
	353-363
	369-379
	417-427
	433-443
	481-491
	497-507

7. 4 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 7-8 DC Measuring Unit Allocation Table

DCMPX = D1, B18 16DC MODE		
Driver channel	I/O channel	DC channel
PD1-11A1 PD17-27A1 PD65-75A1 PD81-91A1	P33-43A P49-59A P97-107A P113-123A	DC1 DC2 DC3 DC4
PD129-139A1 PD145-155A1 PD193-203A1 PD209-219A1	P161-171A P177-187A P225-235A P241-251A	DC5 DC6 DC7 DC8
PD257-267A1 PD273-283A1 PD321-331A1 PD337-347A1 PD385-395A1 PD401-411A1 PD449-459A1 PD465-475A1	P289-299A P305-315A P353-363A P369-379A P417-427A P433-443A P481-491A P497-507A	DC9 DC10 DC11 DC12 DC13 DC14 DC15 DC16

8. 2880CH Test Head

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 2880CH test head.

8. 1 Test Head Configurations and Available Child Pins

2880CH test head has two configurations, Full and Half. The child pins shown in the table below can be used for each configuration.

Test head configuration		Child			
2880CH	Full	A1, A2, A3, A4	B1, B2, B3, B4	C1, C2, C3, C4	D1, D2, D3, D4
		E1, E2, E3, E4	F1, F2, F3, F4	G1, G2, G3, G4	H1, H2, H3, H4
	Half	A1, A2, A3, A4	B1, B2, B3, B4	C1, C2, C3, C4	D1, D2, D3, D4

8. 2 DC Dedicated Channels and Driver Dedicated Channels

DC dedicated channels and driver dedicated channels in the table below are provided.

Channel type	Channel
DC dedicated channel	31, 32, 95, 96
Driver dedicated channel	157 to 160, 221 to 224

8. 3 Tester Channel Allocation Table (1)

Table 8-1 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test				
PD1-24A1, PD31A1, PD32A1, PD65-88A1, PD95A1, PD96A1, PD157-160A1, PD221-224A1	P33-50A1, P97-114A1, P161-178A1, P225-242A1	1				
PD1-24A3, PD31A3, PD32A3, PD65-88A3, PD95A3, PD96A3, PD157-160A3, PD221-224A3	P33-50A3, P97-114A3, P161-178A3, P225-242A3	----	2	4	8	16
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - -	A - - - - - -	A B - - - -	A B C D - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
32	33-48, 97-112
36	33-50, 97-114
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 8-6](#).

8. 4 Tester Channel Allocation Table (2)

Table 8-2 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1, PD31A1, PD32A1, PD65-88A1, PD95A1, PD96A1, PD157-160A1	P33-50A1, P97-114A1	4	8	16	32
PD1-24A2, PD31A2, PD32A2, PD65-88A2, PD95A2, PD96A2, PD221-224A1	P161-178A1, P225-242A1				
PD1-24A3, PD31A3, PD32A3, PD65-88A3, PD95A3, PD96A3, PD157-160A3	P33-50A3, P97-114A3				
PD1-24A4, PD31A4, PD32A4, PD65-88A4, PD95A4, PD96A4, PD221-224A3	P161-178A3, P225-242A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - - -	A B - - - - - -	A B C D - - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242

➔ For more information on DC measurement unit allocation, refer to [Table 8-6](#).

8. 5 Tester Channel Allocation Table (3)

Table 8-3 PIN MODE = A 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test							
PD1-24A1, PD31A1, PD32A1, PD157-158A1	P33-50A1	4	8	16	32	64			
PD65-88A1, PD95A1, PD96A1, PD159-160A1,	P97-114A1								
PD1-24A3, PD31A3, PD32A3, PD157-158A3	P33-50A3								
PD65-88A3, PD95A3, PD96A3, PD159-160A3,	P97-114A3								
PD1-24A2, PD31A2, PD32A2, PD221-222A1	P161-178A1	----	8	16	32	64			
PD65-88A2, PD95A2, PD96A2, PD223-224A1	P225-242A1								
PD1-24A4, PD31A4, PD32A4, PD221-222A3	P161-178A3								
PD65-88A4, PD95A4, PD96A4, PD223-224A3	P225-242A3								
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A	A	A	A	A
		-	-	B	B	B	B	B	B
		-	-	-	-	C	C	C	C
		-	-	-	-	D	D	D	D
		-	-	-	-	-	-	E	E
		-	-	-	-	-	-	F	F
		-	-	-	-	-	-	G	G
		-	-	-	-	-	-	H	H

8. 5 Tester Channel Allocation Table (3)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242

➔ For more information on DC measurement unit allocation, refer to [Table 8-7](#).

8. 6 Tester Channel Allocation Table (4)

Table 8-4 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-24A1, PD31A1, PD32A1, PD157-158A1	P33-40A1, 49A1	128	----	----
PD1-24A1, PD31A1, PD32A1, PD157-158A1	P41-48A1, 50A1			
PD65-88A1, PD95A1, PD96A1, PD159-160A1	P97-104A1, 113A1			
PD65-88A1, PD95A1, PD96A1, PD159-160A1	P105-112A1, 114A1			
PD1-24A2, PD31A2, PD32A2, PD221-222A1	P161-168A1, 177A1			
PD1-24A2, PD31A2, PD32A2, PD221-222A1	P169-176A1, 178A1			
PD65-88A2, PD95A2, PD96A2, PD223-224A1	P225-232A1, 241A1			
PD65-88A2, PD95A2, PD96A2, PD223-224A1	P233-240A1, 242A1			
PD1-24A3, PD31A3, PD32A3, PD157-158A3	P33-40A3, 49A3			
PD1-24A3, PD31A3, PD32A3, PD157-158A3	P41-48A3, 50A3			
PD65-88A3, PD95A3, PD96A3, PD159-160A3	P97-104A3, 113A3			
PD65-88A3, PD95A3, PD96A3, PD159-160A3	P105-112A3, 114A3			
PD1-24A4, PD31A4, PD32A4, PD221-222A3	P161-168A3, 177A3			
PD1-24A4, PD31A4, PD32A4, PD221-222A3	P169-176A3, 178A3			
PD65-88A4, PD95A4, PD96A4, PD223-224A3	P225-232A3, 241A3			
PD65-88A4, PD95A4, PD96A4, PD223-224A3	P233-240A3, 242A3			
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	----	----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
	161
	169
	225
	233
4	33-36
	41-44
	97-100
	105-108
	161-164
	169-172
	225-228
	233-236
8	33-40
	41-48
	97-104
	105-112
	161-168
	169-176
	225-232
	233-240
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
	161-168, 177
	169-176, 178
	225-232, 241
	233-240, 242

➔ For more information on DC measurement unit allocation, refer to [Table 8-7](#).

8. 7 Tester Channel Allocation Table (5)

Table 8-5 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test		
PD1-12A1, PD31A1	P33-40A1, 49A1	128	----	----
PD13-24A1, PD32A1	P41-48A1, 50A1			
PD65-76A1, PD95A1	P97-104A1, 113A1			
PD77-88A1, PD96A1	P105-112A1, 114A1			
PD1-12A2, PD31A2	P161-168A1, 177A1			
PD13-24A2, PD32A2	P169-176A1, 178A1			
PD65-76A2, PD95A2	P225-232A1, 241A1			
PD77-88A2, PD96A2	P233-240A1, 242A1			
PD1-12A3, PD31A3	P33-40A3, 49A3			
PD13-24A3, PD32A3	P41-48A3, 50A3			
PD65-76A3, PD95A3	P97-104A3, 113A3			
PD77-88A3, PD96A3	P105-112A3, 114A3			
PD1-12A4, PD31A4	P161-168A3, 177A3			
PD13-24A4, PD32A4	P169-176A3, 178A3			
PD65-76A4, PD95A4	P225-232A3, 241A3			
PD77-88A4, PD96A4	P233-240A3, 242A3			
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	----	----

8. 7 Tester Channel Allocation Table (5)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
	161
	169
	225
	233
4	33-36
	41-44
	97-100
	105-108
	161-164
	169-172
	225-228
	233-236
8	33-40
	41-48
	97-104
	105-112
	161-168
	169-176
	225-232
	233-240
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
	161-168, 177
	169-176, 178
	225-232, 241
	233-240, 242

➔ For more information on DC measurement unit allocation, refer to [Table 8-7](#).

8. 8 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 8-6 DC Measuring Unit Allocation Table (1)

DCMPX = D1, B36 32DC/64DC MODE										
Child	Tester channel									
	1-24, 31, 32	65-88, 95, 96	157, 158	159, 160	221, 222	223, 224	33-50	97-114	161-178	225-242
A1	DC1	DC1	DC2	DC1	DC2	DC1	DC2	DC1	DC2	DC2
A2	DC2	DC2	DC3	DC3	DC4	DC3	DC4	DC3	DC4	DC4
A3	DC3	DC3	DC4	DC4	DC5	DC5	DC6	DC5	DC6	DC6
A4	DC4	DC4	DC5	DC5	DC6	DC6	DC7	DC7	DC8	DC8
B1	DC5	DC5	DC6	DC6	DC7	DC7	DC8	DC7	DC8	DC8
B2	DC6	DC6	DC7	DC7	DC8	DC8	DC9	DC9	DC10	DC10
B3	DC7	DC7	DC8	DC8	DC9	DC9	DC10	DC9	DC10	DC10
B4	DC8	DC8	DC9	DC9	DC10	DC10	DC11	DC11	DC12	DC12
C1	DC33	DC33	DC34	DC34	DC35	DC35	DC36	DC35	DC36	DC36
C2	DC34	DC34	DC35	DC35	DC36	DC36	DC37	DC37	DC38	DC38
C3	DC35	DC35	DC36	DC36	DC37	DC37	DC38	DC37	DC38	DC38
C4	DC36	DC36	DC37	DC37	DC38	DC38	DC39	DC39	DC40	DC40
D1	DC37	DC37	DC38	DC38	DC39	DC39	DC40	DC39	DC40	DC40
D2	DC38	DC38	DC39	DC39	DC40	DC40	DC41	DC41	DC42	DC42
D3	DC39	DC39	DC40	DC40	DC41	DC41	DC42	DC41	DC42	DC42
D4	DC40	DC40	DC41	DC41	DC42	DC42	DC43	DC43	DC44	DC44
E1	DC9	DC9	DC10	DC10	DC11	DC11	DC12	DC11	DC12	DC12
E2	DC10	DC10	DC11	DC11	DC12	DC12	DC13	DC13	DC14	DC14
E3	DC11	DC11	DC12	DC12	DC13	DC13	DC14	DC13	DC14	DC14
E4	DC12	DC12	DC13	DC13	DC14	DC14	DC15	DC15	DC16	DC16
F1	DC13	DC13	DC14	DC14	DC15	DC15	DC16	DC15	DC16	DC16
F2	DC14	DC14	DC15	DC15	DC16	DC16	DC17	DC17	DC18	DC18
F3	DC15	DC15	DC16	DC16	DC17	DC17	DC18	DC17	DC18	DC18
F4	DC16	DC16	DC17	DC17	DC18	DC18	DC19	DC19	DC20	DC20
G1	DC41	DC41	DC42	DC42	DC43	DC43	DC44	DC43	DC44	DC44
G2	DC42	DC42	DC43	DC43	DC44	DC44	DC45	DC45	DC46	DC46
G3	DC43	DC43	DC44	DC44	DC45	DC45	DC46	DC45	DC46	DC46
G4	DC44	DC44	DC45	DC45	DC46	DC46	DC47	DC47	DC48	DC48
H1	DC45	DC45	DC46	DC46	DC47	DC47	DC48	DC47	DC48	DC48
H2	DC46	DC46	DC47	DC47	DC48	DC48	DC49	DC49	DC50	DC50
H3	DC47	DC47	DC48	DC48	DC49	DC49	DC50	DC49	DC50	DC50
H4	DC48	DC48	DC49	DC49	DC50	DC50	DC51	DC51	DC52	DC52

Table 8-7 DC Measuring Unit Allocation Table (2)

DCMPX = D1, B18 32DC/64DC MODE										
Child	Tester channel									
	1-24, 31, 32	65-88, 95, 96	157, 158	159, 160	221, 222	223, 224	33-50	97-114	161-178	225-242
A1	DC1	DC17	DC1	DC17	DC2	DC18	DC1	DC17	DC2	DC18
A2	DC2	DC18	-----							
A3	DC3	DC19	DC3	DC19	DC4	DC20	DC3	DC19	DC4	DC20
A4	DC4	DC20	-----							
B1	DC5	DC21	DC5	DC21	DC6	DC22	DC5	DC21	DC6	DC22
B2	DC6	DC22	-----							
B3	DC7	DC23	DC7	DC23	DC8	DC24	DC7	DC23	DC8	DC24
B4	DC8	DC24	-----							
C1	DC33	DC49	DC33	DC49	DC34	DC50	DC33	DC49	DC34	DC50
C2	DC34	DC50	-----							
C3	DC35	DC51	DC35	DC51	DC36	DC52	DC35	DC51	DC36	DC52
C4	DC36	DC52	-----							
D1	DC37	DC53	DC37	DC53	DC38	DC54	DC37	DC53	DC38	DC54
D2	DC38	DC54	-----							
D3	DC39	DC55	DC39	DC55	DC40	DC56	DC39	DC55	DC40	DC56
D4	DC40	DC56	-----							
E1	DC9	DC25	DC9	DC25	DC10	DC26	DC9	DC25	DC10	DC26
E2	DC10	DC26	-----							
E3	DC11	DC27	DC11	DC27	DC12	DC28	DC11	DC27	DC12	DC28
E4	DC12	DC28	-----							
F1	DC13	DC29	DC13	DC29	DC14	DC30	DC13	DC29	DC14	DC30
F2	DC14	DC30	-----							
F3	DC15	DC31	DC15	DC31	DC16	DC32	DC15	DC31	DC16	DC32
F4	DC16	DC32	-----							
G1	DC41	DC57	DC41	DC57	DC42	DC58	DC41	DC57	DC42	DC58
G2	DC42	DC58	-----							
G3	DC43	DC59	DC43	DC59	DC44	DC60	DC43	DC59	DC44	DC60
G4	DC44	DC60	-----							
H1	DC45	DC61	DC45	DC61	DC46	DC62	DC45	DC61	DC46	DC62
H2	DC46	DC62	-----							
H3	DC47	DC63	DC47	DC63	DC48	DC64	DC47	DC63	DC48	DC64
H4	DC48	DC64	-----							

9. 1344CH Test Head

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 1344CH test head.

9. 1 Test Head Configurations, Available Child Pins, and Channel Numbers

Test head configuration		Child	Channel number
1344CH	Full	A, B, C, D, E, F, G, H	1 to 248
	Half	A, B, C, D, -, -, -, -	1 to 248
	Quarter	A, B, C, D, -, -, -, -	1 to 120

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

51 to 56, 115 to 120, 179 to 184, 243 to 248

9. 2 Tester Channel Allocation Table (1)

Table 9-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-21A, PD65-85A, PD129-149A, PD193-213A	P33-56A, P97-120A, P161-184A, P225-248A	1	2	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - -	A B - - - -	A B C D - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
24	33-56
32	33-48, 97-112
36	33-50, 97-114
48	33-56, 97-120
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242
96	33-56, 97-120, 161-184, 225-248

➔ For more information on DC measurement unit allocation, refer to [Table 9-5](#).

9. 3 Tester Channel Allocation Table (2)

Table 9-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-21A, PD65-85A	P33-56A, P97-120A	2	4	8	16
PD129-149A, PD193-213A	P161-184A, P225-248A				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
24	33-56
	161-184
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242
48	33-56, 97-120
	161-184, 225-248

➔ For more information on DC measurement unit allocation, refer to [Table 9-6](#).

9. 4 Tester Channel Allocation Table (3)

Table 9-3 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-21A	P33-56A	4	8	16	32
PD65-85A	P97-120A				
PD129-149A	P161-184A				
PD193-213A	P225-248A				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

9. 4 Tester Channel Allocation Table (3)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242
24	33-56
	97-120
	161-184
	225-248

➔ For more information on DC measurement unit allocation, refer to [Table 9-7](#).

9. 5 Tester Channel Allocation Table (4)

Table 9-4 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-21A	P33-56A	2	4	8	16
PD65-85A	P97-120A				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
24	33-56
	97-120

➔ For more information on DC measurement unit allocation, refer to [Table 9-8](#).

9. 6 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 9-5 DC Measuring Unit Allocation Table (1)

DCMPX = D1, B96 16DC MODE								
Child	Tester channel							
	1-21	65-85	129-149	193-213	33-56	97-120	161-184	225-248
A	DC1							
B	DC3							
C	DC5							
D	DC7							
E	DC9							
F	DC11							
G	DC13							
H	DC15							

Table 9-6 DC Measuring Unit Allocation Table (2)

DCMPX = D1, B36 16DC MODE								
Child	Tester channel							
	1-21	65-85	129-149	193-213	33-56	97-120	161-184	225-248
A	DC1		DC2		DC1		DC2	
B	DC3		DC4		DC3		DC4	
C	DC5		DC6		DC5		DC6	
D	DC7		DC8		DC7		DC8	
E	DC9		DC10		DC9		DC10	
F	DC11		DC12		DC11		DC12	
G	DC13		DC14		DC13		DC14	
H	DC15		DC16		DC15		DC16	

Table 9-7 DC Measuring Unit Allocation Table (3)

DCMPX = D1, B18 16DC/32DC MODE								
Child	Tester channel							
	1-21	65-85	129-149	193-213	33-56	97-120	161-184	225-248
A	DC1	DC17	DC2	DC18	DC1	DC17	DC2	DC18
B	DC3	DC19	DC4	DC20	DC3	DC19	DC4	DC20
C	DC5	DC21	DC6	DC22	DC5	DC21	DC6	DC22
D	DC7	DC23	DC8	DC24	DC7	DC23	DC8	DC24
E	DC9	DC25	DC10	DC26	DC9	DC25	DC10	DC26
F	DC11	DC27	DC12	DC28	DC11	DC27	DC12	DC28
G	DC13	DC29	DC14	DC30	DC13	DC29	DC14	DC30
H	DC15	DC31	DC16	DC32	DC15	DC31	DC16	DC32

Table 9-8 DC Measuring Unit Allocation Table (4)

DCMPX = D1, B9 16DC MODE				
Child	Tester channel			
	1-21	65-85	33-56	97-120
A	DC1	DC2	DC1	DC2
B	DC3	DC4	DC3	DC4
C	DC5	DC6	DC5	DC6
D	DC7	DC8	DC7	DC8
E	DC9	DC10	DC9	DC10
F	DC11	DC12	DC11	DC12
G	DC13	DC14	DC13	DC14
H	DC15	DC16	DC15	DC16

10. 3712(1280IO)CH or 3456(1280IO)CH Test Head

This chapter shows the tester channel allocation table used for parallel test of multiple DUTs by 3712(1280IO)CH or 3456(1280IO)CH test head.

10. 1 Test Head Configurations and Available Child Pins

3712(1280IO)CH or 3456(1280IO)CH test head has three configurations, Full, Half, and Quarter, and the child pins shown in the table below can be used.

Test head configuration		Child	
3712(1280IO)CH or 3456(1280IO)CH	Full	A1, A2, A3, A4, A5, A6, A7, A8	B1, B2, B3, B4, B5, B6, B7, B8
		C1, C2, C3, C4, C5, C6, C7, C8	D1, D2, D3, D4, D5, D6, D7, D8
		E1, E2, E3, E4, E5, E6, E7, E8	F1, F2, F3, F4, F5, F6, F7, F8
		G1, G2, G3, G4, G5, G6, G7, G8	H1, H2, H3, H4, H5, H6, H7, H8
	Half	A1, A2, A3, A4, A5, A6, A7, A8	B1, B2, B3, B4, B5, B6, B7, B8
		C1, C2, C3, C4, C5, C6, C7, C8	D1, D2, D3, D4, D5, D6, D7, D8
		-----	-----
		-----	-----
	Quarter	A1, A2, A3, A4, A5, A6, A7, A8	B1, B2, B3, B4, B5, B6, B7, B8
		-----	-----
		-----	-----
		-----	-----

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

51, 52, 115, 116

10. 2 DC Dedicated Channels and Level Driver Channels

DC dedicated channels and LV (Level) driver channels in the table below are provided.

Channel type	Channel
DC dedicated channel	31, 32
LV driver channel	67, 68, 69, 70



Limitation

DC measuring units cannot be connected to the LV driver channel 67, 68, 69, and 70.

10. 3 Tester Channel Allocation Table (1)

Table 10-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1, P97-116A1	1	2	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
20	33-52
32	33-48, 97-112
36	33-50, 97-114
40	33-52, 97-116

10. 4 Tester Channel Allocation Table (2)

Table 10-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1, P97-116A1	2	4	8	16
PD1-32A3, PD65-70A3	P33-52A3, P97-116A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
20	33-52
32	33-48, 97-112
36	33-50, 97-114
40	33-52, 97-116

10. 5 Tester Channel Allocation Table (3)

Table 10-3 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1	2	4	8	16
PD1-32A2, PD65-70A2	P97-116A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
20	33-52
	97-116

10. 6 Tester Channel Allocation Table (4)

Table 10-4 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1, P97-116A1	4	8	-----	-----
PD1-32A5, PD65-70A5	P33-52A5, P97-116A5				
Child names to be used (When child C, E and G are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A		
		-	-		
		C	C		
		-	-		
		-	E	-----	-----
		-	-		
		-	G		
		-	-		

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
20	33-52
32	33-48, 97-112
36	33-50, 97-114
40	33-52, 97-116

10. 7 Tester Channel Allocation Table (5)

Table 10-5 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1, P97-116A1	4	8	16	32
PD1-32A3, PD65-70A3	P33-52A3, P97-116A3				
PD1-32A5, PD65-70A5	P33-52A5, P97-116A5				
PD1-32A7, PD65-70A7	P33-52A7, P97-116A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
20	33-52
32	33-48, 97-112
36	33-50, 97-114
40	33-52, 97-116

10. 8 Tester Channel Allocation Table (6)

Table 10-6 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1	4	8	16	32
PD1-32A2, PD65-70A2	P97-116A1				
PD1-32A3, PD65-70A3	P33-52A3				
PD1-32A4, PD65-70A4	P97-116A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
20	33-52
	97-116

10. 9 Tester Channel Allocation Table (7)

Table 10-7 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1	8	16	-----	-----
PD1-32A2, PD65-70A2	P97-116A1				
PD1-32A5, PD65-70A5	P33-52A5				
PD1-32A6, PD65-70A6	P97-116A5				
Child names to be used (When child C, E and G are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A		
		-	-		
		C	C		
		-	-		
		-	E	-----	-----
		-	-		
		-	G		
		-	-		

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
20	33-52
	97-116

10.10 Tester Channel Allocation Table (8)

Table 10-8 PIN MODE = A 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-52A1	8	16	32	64
PD1-32A2, PD65-70A2	P97-116A1				
PD1-32A3, PD65-70A3	P33-52A3				
PD1-32A4, PD65-70A4	P97-116A3				
PD1-32A5, PD65-70A5	P33-52A5				
PD1-32A6, PD65-70A6	P97-116A5				
PD1-32A7, PD65-70A7	P33-52A7				
PD1-32A8, PD65-70A8	P97-116A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - - -	A B - - - - - -	A B C D - - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
20	33-52
	97-116

10.11 Tester Channel Allocation Table (9)

Table 10-9 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-12A1, PD25-27A1, PD31A1, PD65A1, PD67A1, PD69A1	P33-40A1, P49A1, P51A1	8	16	32	64
PD13-24A1, PD28-30A1, PD32A1, PD66A1, PD68A1, PD70A1	P41-48A1, P50A1, P52A1				
PD1-12A2, PD25-27A2, PD31A2, PD65A2, PD67A2, PD69A2	P97-104A1, P113A1, P115A1				
PD13-24A2, PD28-30A2, PD32A2, PD66A2, PD68A2, PD70A2	P105-112A1, P114A1, P116A1				
PD1-12A3, PD25-27A3, PD31A3, PD65A3, PD67A3, PD69A3	P33-40A3, P49A3, P51A3				
PD13-24A3, PD28-30A3, PD32A3, PD66A3, PD68A3, PD70A3	P41-48A3, P50A3, P52A3				
PD1-12A4, PD25-27A4, PD31A4, PD65A4, PD67A4, PD69A4	P97-104A3, P113A3, P115A3				
PD13-24A4, PD28-30A4, PD32A4, PD66A4, PD68A4, PD70A4	P105-112A3, P114A3, P116A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - - -	A B - - - - - -	A B C D - - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

10.12 Tester Channel Allocation Table (10)

Table 10-10 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-12A1, PD25-27A1, PD31A1, PD65A1, PD67A1, PD69A1	P33-40A1, P49A1, P51A1	64	128	-----	-----
PD13-24A1, PD28-30A1, PD32A1, PD66A1, PD68A1, PD70A1	P41-48A1, P50A1, P52A1				
PD1-12A2, PD25-27A2, PD31A2, PD65A2, PD67A2, PD69A2	P97-104A1, P113A1, P115A1				
PD13-24A2, PD28-30A2, PD32A2, PD66A2, PD68A2, PD70A2	P105-112A1, P114A1, P116A1				
PD1-12A3, PD25-27A3, PD31A3, PD65A3, PD67A3, PD69A3	P33-40A3, P49A3, P51A3				
PD13-24A3, PD28-30A3, PD32A3, PD66A3, PD68A3, PD70A3	P41-48A3, P50A3, P52A3				
PD1-12A4, PD25-27A4, PD31A4, PD65A4, PD67A4, PD69A4	P97-104A3, P113A3, P115A3				
PD13-24A4, PD28-30A4, PD32A4, PD66A4, PD68A4, PD70A4	P105-112A3, P114A3, P116A3				
PD1-12A5, PD25-27A5, PD31A5, PD65A5, PD67A5, PD69A5	P33-40A5, P49A5, P51A5				
PD13-24A5, PD28-30A5, PD32A5, PD66A5, PD68A5, PD70A5	P41-48A5, P50A5, P52A5				
PD1-12A6, PD25-27A6, PD31A6, PD65A6, PD67A6, PD69A6	P97-104A5, P113A5, P115A5				
PD13-24A6, PD28-30A6, PD32A6, PD66A6, PD68A6, PD70A6	P105-112A5, P114A5, P116A5				
PD1-12A7, PD25-27A7, PD31A7, PD65A7, PD67A7, PD69A7	P33-40A7, P49A7, P51A7				
PD13-24A7, PD28-30A7, PD32A7, PD66A7, PD68A7, PD70A7	P41-48A7, P50A7, P52A7				
PD1-12A8, PD25-27A8, PD31A8, PD65A8, PD67A8, PD69A8	P97-104A7, P113A7, P115A7				
PD13-24A8, PD28-30A8, PD32A8, PD66A8, PD68A8, PD70A8	P105-112A7, P114A7, P116A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D - - - -	A B C D E F G H	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

10.13 Tester Channel Allocation Table (11)

Table 10-11 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-40A1, P49A1, P51A1	128	-----	-----	-----
PD1-32A1, PD65-70A1	P41-48A1, P50A1, P52A1				
PD1-32A2, PD65-70A2	P97-104A1, P113A1, P115A1				
PD1-32A2, PD65-70A2	P105-112A1, P114A1, P116A1				
PD1-32A3, PD65-70A3	P33-40A3, P49A3, P51A3				
PD1-32A3, PD65-70A3	P41-48A3, P50A3, P52A3				
PD1-32A4, PD65-70A4	P97-104A3, P113A3, P115A3				
PD1-32A4, PD65-70A4	P105-112A3, P114A3, P116A3				
PD1-32A5, PD65-70A5	P33-40A5, P49A5, P51A5				
PD1-32A5, PD65-70A5	P41-48A5, P50A5, P52A5				
PD1-32A6, PD65-70A6	P97-104A5, P113A5, P115A5				
PD1-32A6, PD65-70A6	P105-112A5, P114A5, P116A5				
PD1-32A7, PD65-70A7	P33-40A7, P49A7, P51A7				
PD1-32A7, PD65-70A7	P41-48A7, P50A7, P52A7				
PD1-32A8, PD65-70A8	P97-104A7, P113A7, P115A7				
PD1-32A8, PD65-70A8	P105-112A7, P114A7, P116A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

10.14 Tester Channel Allocation Table (12)

Table 10-12 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-40A1, P49A1, P51A1	8	16	32	64
PD1-32A2, PD65-70A2	P41-48A1, P50A1, P52A1				
PD1-32A5, PD65-70A5	P97-104A1, P113A1, P115A1				
PD1-32A6, PD65-70A6	P105-112A1, P114A1, P116A1				
PD1-32A3, PD65-70A3	P33-40A3, P49A3, P51A3				
PD1-32A4, PD65-70A4	P41-48A3, P50A3, P52A3				
PD1-32A7, PD65-70A7	P97-104A3, P113A3, P115A3				
PD1-32A8, PD65-70A8	P105-112A3, P114A3, P116A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - - -	A B - - - - - -	A B C D - - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

10.15 Tester Channel Allocation Table (13)

Table 10-13 PIN MODE = C 1 Child: 32

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-36A1, P49A1	256	-----	-----	-----
PD1-32A1, PD65-70A1	P37-40A1, P51A1				
PD1-32A1, PD65-70A1	P41-44A1, P50A1				
PD1-32A1, PD65-70A1	P45-48A1, P52A1				
PD1-32A2, PD65-70A2	P97-100A1, P113A1				
PD1-32A2, PD65-70A2	P101-104A1, P115A1				
PD1-32A2, PD65-70A2	P105-108A1, P114A1				
PD1-32A2, PD65-70A2	P109-112A1, P116A1				
PD1-32A3, PD65-70A3	P33-36A3, P49A3				
PD1-32A3, PD65-70A3	P37-40A3, P51A3				
PD1-32A3, PD65-70A3	P41-44A3, P50A3				
PD1-32A3, PD65-70A3	P45-48A3, P52A3				
PD1-32A4, PD65-70A4	P97-100A3, P113A3				
PD1-32A4, PD65-70A4	P101-104A3, P115A3				
PD1-32A4, PD65-70A4	P105-108A3, P114A3				
PD1-32A4, PD65-70A4	P109-112A3, P116A3				
PD1-32A5, PD65-70A5	P33-36A5, P49A5				
PD1-32A5, PD65-70A5	P37-40A5, P51A5				
PD1-32A5, PD65-70A5	P41-44A5, P50A5				
PD1-32A5, PD65-70A5	P45-48A5, P52A5				
PD1-32A6, PD65-70A6	P97-100A5, P113A5				
PD1-32A6, PD65-70A6	P101-104A5, P115A5				
PD1-32A6, PD65-70A6	P105-108A5, P114A5				
PD1-32A6, PD65-70A6	P109-112A5, P116A5				
PD1-32A7, PD65-70A7	P33-36A7, P49A7				
PD1-32A7, PD65-70A7	P37-40A7, P51A7				
PD1-32A7, PD65-70A7	P41-44A7, P50A7				
PD1-32A7, PD65-70A7	P45-48A7, P52A7				
PD1-32A8, PD65-70A8	P97-100A7, P113A7				
PD1-32A8, PD65-70A8	P101-104A7, P115A7				
PD1-32A8, PD65-70A8	P105-108A7, P114A7				
PD1-32A8, PD65-70A8	P109-112A7, P116A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	37
	41
	45
	97
	101
	105
	109
4	33-36
	37-40
	41-44
	45-48
	97-100
	101-104
	105-108
	109-112
5	33-36, 49
	37-40, 51
	41-44, 50
	45-48, 52
	97-100, 113
	101-104, 115
	105-108, 114
	109-112, 116

10.16 Tester Channel Allocation Table (14)

Table 10-14 PIN MODE = C 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1, PD65-70A1	P33-36A1, P49A1	128	-----	-----	-----
PD1-32A1, PD65-70A1	P37-40A1, P51A1				
PD1-32A2, PD65-70A2	P41-44A1, P50A1				
PD1-32A2, PD65-70A2	P45-48A1, P52A1				
PD1-32A5, PD65-70A5	P97-100A1, P113A1				
PD1-32A5, PD65-70A5	P101-104A1, P115A1				
PD1-32A6, PD65-70A6	P105-108A1, P114A1				
PD1-32A6, PD65-70A6	P109-112A1, P116A1				
PD1-32A3, PD65-70A3	P33-36A3, P49A3				
PD1-32A3, PD65-70A3	P37-40A3, P51A3				
PD1-32A4, PD65-70A4	P41-44A3, P50A3				
PD1-32A4, PD65-70A4	P45-48A3, P52A3				
PD1-32A7, PD65-70A7	P97-100A3, P113A3				
PD1-32A7, PD65-70A7	P101-104A3, P115A3				
PD1-32A8, PD65-70A8	P105-108A3, P114A3				
PD1-32A8, PD65-70A8	P109-112A3, P116A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	37
	41
	45
	97
	101
	105
	109
4	33-36
	37-40
	41-44
	45-48
	97-100
	101-104
	105-108
	109-112
5	33-36, 49
	37-40, 51
	41-44, 50
	45-48, 52
	97-100, 113
	101-104, 115
	105-108, 114
	109-112, 116

10.17 Tester Channel Allocation Table (15)

Table 10-15 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1 PD65-70A1	P33-52A1 P97-116A1	4	8	16	-----
PD1-32A5 PD65-70A5	P33-52A5 P97-116A5				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	-----
		B	B	B	
		-	C	C	
		-	D	D	
		-	-	E	
		-	-	F	
		-	-	G	
		-	-	H	

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
20	33-52
32	33-48, 97-112
36	33-50, 97-114
40	33-52, 97-116

10.18 Tester Channel Allocation Table (16)

Table 10-16 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1 PD65-70A1	P33-52A1	8	16	32	-----
PD1-32A2 PD65-70A2	P97-116A1				
PD1-32A5 PD65-70A5	P33-52A5				
PD1-32A6 PD65-70A6	P97-116A5				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	-----
		B	B	B	
		-	C	C	
		-	D	D	
		-	-	E	
		-	-	F	
		-	-	G	
		-	-	H	

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
20	33-52
	97-116

10.19 Tester Channel Allocation Table (17)

Table 10-17 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1 PD65-70A1	P33-40A1 P49A1 P51A1	16	32	64	-----
PD1-32A2 PD65-70A2	P41-48A1 P50A1 P52A1				
PD1-32A3 PD65-70A3	P97-104A1 P113A1 P115A1				
PD1-32A4 PD65-70A4	P105-112A1 P114A1 P116A1				
PD1-32A5 PD65-70A5	P33-40A5 P49A5 P51A5				
PD1-32A6 PD65-70A6	P41-48A5 P50A5 P52A5				
PD1-32A7 PD65-70A7	P97-104A5 P113A5 P115A5				
PD1-32A8 PD65-70A8	P105-112A5 P114A5 P116A5				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B - - - - -	A B C D - - -	A B C D E F G H	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
4	33-36
	41-44
	97-100
	105-108
8	33-40
	41-48
	97-104
	105-112
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
10	33-40, 49, 51
	41-48, 50, 52
	97-104, 113, 115
	105-112, 114, 116

10.20 Tester Channel Allocation Table (18)

Table 10-18 PIN MODE = B 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-32A1 PD65-70A1	P33-40A1 P49A1 P51A1	16	32	64	-----
PD1-32A2 PD65-70A2	P41-48A1 P50A1 P52A1				
PD1-32A3 PD65-70A3	P33-40A3 P49A3 P51A3				
PD1-32A4 PD65-70A4	P41-48A3 P50A3 P52A3				
PD1-32A5 PD65-70A5	P33-40A5 P49A5 P51A5				
PD1-32A6 PD65-70A6	P41-48A5 P50A5 P52A5				
PD1-32A7 PD65-70A7	P33-40A7 P49A7 P51A7				
PD1-32A8 PD65-70A8	P41-48A7 P50A7 P52A7				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B - - - - -	A B C D - - -	A B C D E F G H	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
4	33-36
	41-44
8	33-40
	41-48
9	33-40, 49
	41-48, 50
10	33-40, 49, 51
	41-48, 50, 52

10.21 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are fixed as shown in the table below.

Table 10-19 DC Measuring Unit Allocation Table

Child	Tester channel				
	1-32	65, 66	67-70	33-52	97-116
A1	DC1		-----	DC1	DC3
A2	DC3		-----	-----	-----
A3	DC2		-----	DC2	DC4
A4	DC4		-----	-----	-----
A5	DC5		-----	DC5	DC7
A6	DC7		-----	-----	-----
A7	DC6		-----	DC6	DC8
A8	DC8		-----	-----	-----
B1	DC17		-----	DC17	DC19
B2	DC19		-----	-----	-----
B3	DC18		-----	DC18	DC20
B4	DC20		-----	-----	-----
B5	DC21		-----	DC21	DC23
B6	DC23		-----	-----	-----
B7	DC22		-----	DC22	DC24
B8	DC24		-----	-----	-----
C1	DC9		-----	DC9	DC11
C2	DC11		-----	-----	-----
C3	DC10		-----	DC10	DC12
C4	DC12		-----	-----	-----
C5	DC13		-----	DC13	DC15
C6	DC15		-----	-----	-----
C7	DC14		-----	DC14	DC16
C8	DC16		-----	-----	-----
D1	DC25		-----	DC25	DC27
D2	DC27		-----	-----	-----
D3	DC26		-----	DC26	DC28
D4	DC28		-----	-----	-----
D5	DC29		-----	DC29	DC31
D6	DC31		-----	-----	-----
D7	DC30		-----	DC30	DC32
D8	DC32		-----	-----	-----
E1	DC33		-----	DC33	DC35
E2	DC35		-----	-----	-----
E3	DC34		-----	DC34	DC36
E4	DC36		-----	-----	-----
E5	DC37		-----	DC37	DC39
E6	DC39		-----	-----	-----
E7	DC38		-----	DC38	DC40

10. 21 DC Measuring Unit Allocation Table

Child	Tester channel				
	1-32	65, 66	67-70	33-52	97-116
E8	DC40		-----	-----	-----
F1	DC49		-----	DC49	DC51
F2	DC51		-----	-----	-----
F3	DC50		-----	DC50	DC52
F4	DC52		-----	-----	-----
F5	DC53		-----	DC53	DC55
F6	DC55		-----	-----	-----
F7	DC54		-----	DC54	DC56
F8	DC56		-----	-----	-----
G1	DC41		-----	DC41	DC43
G2	DC43		-----	-----	-----
G3	DC42		-----	DC42	DC44
G4	DC44		-----	-----	-----
G5	DC45		-----	DC45	DC47
G6	DC47		-----	-----	-----
G7	DC46		-----	DC46	DC48
G8	DC48		-----	-----	-----
H1	DC57		-----	DC57	DC59
H2	DC59		-----	-----	-----
H3	DC58		-----	DC58	DC60
H4	DC60		-----	-----	-----
H5	DC61		-----	DC61	DC63
H6	DC63		-----	-----	-----
H7	DC62		-----	DC62	DC64
H8	DC64		-----	-----	-----

11. 3072CH Test Head

This chapter shows the tester channel allocation table used for parallel testing of multiple DUTs with a 3072CH test head.

11. 1 Test Head Configurations, Available Child Pins, and Channel Numbers

3072CH test head has three configurations, Full, Half, and Quarter, and the child pins and channel numbers shown in the table below can be used.

Test head configuration		Child		Channel number
3072CH	Full	A1, A2, A3, A4	B1, B2, B3, B4	1 to 248
		C1, C2, C3, C4	D1, D2, D3, D4	
		E1, E2, E3, E4	F1, F2, F3, F4	
		G1, G2, G3, G4	H1, H2, H3, H4	
	Half	A1, A2, -, -	B1, B2, -, -	1 to 248
		C1, C2, -, -	D1, D2, -, -	
		E1, E2, -, -	F1, F2, -, -	
		G1, G2, -, -	H1, H2, -, -	
	Quarter	A1, -, -, -	B1, -, -, -	1 to 120
		C1, -, -, -	D1, -, -, -	
		E1, -, -, -	F1, -, -, -	
		G1, -, -, -	H1, -, -, -	

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

51 to 56, 115 to 120, 179 to 184, 243 to 248

11. 2 Tester Channel Allocation Table (1)

Table 11-1 PIN MODE = A 1 Child: 1

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1 PD65-88A1	P33-56A1 P97-120A1 P161-184A1 P225-248A1	1	2	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - -	A B - - - - -	A B C D - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
24	33-56
32	33-48, 97-112
36	33-50, 97-114
48	33-56, 97-120
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242
96	33-56, 97-120, 161-184, 225-248

Limitation

The number of bits that exceeds 48 cannot be used in a Quarter configuration test head.

11. 3 Tester Channel Allocation Table (2)

Table 11-2 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1 PD65-88A1	P33-56A1 P97-120A1 P161-184A1 P225-248A1	2	4	8	16
PD1-24A3 PD65-88A3	P33-56A3 P97-120A3 P161-184A3 P225-248A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - -	A B - - - -	A B C D E F G H	

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
24	33-56
32	33-48, 97-112
36	33-50, 97-114
48	33-56, 97-120
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242
96	33-56, 97-120, 161-184, 225-248

Limitation

The number of bits that exceeds 48 cannot be used in a Quarter configuration test head.

11. 4 Tester Channel Allocation Table (3)

Table 11-3 PIN MODE = A 1 Child: 2

Driver Channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1 PD65-88A1	P33-56A1 P97-120A1	2	4	8	16
PD1-24A2 PD65-88A2	P161-184A1 P225-248A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
24	33-56
	161-184
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242
48	33-56, 97-120
	161-184, 225-248

11. 5 Tester Channel Allocation Table (4)

Table 11-4 PIN MODE = A 1 Child: 2

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1	P33-56A1	2	4	8	16
PD65-88A1	P97-120A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
24	33-56
	97-120

11. 6 Tester Channel Allocation Table (5)

Table 11-5 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1 PD65-88A1	P33-56A1 P97-120A1	4	8	16	32
PD1-24A2 PD65-88A2	P161-184A1 P225-248A1				
PD1-24A3 PD65-88A3	P33-56A3 P97-120A3				
PD1-24A4 PD65-88A4	P161-184A3 P225-248A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - -	A B - - - - -	A B C D E F G H	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
24	33-56
	161-184
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242
48	33-56, 97-120
	161-184, 225-248

11. 7 Tester Channel Allocation Table (6)

Table 11-6 PIN MODE = A 1 Child: 4

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1	P33-56A1	4	8	16	32
PD65-88A1	P97-120A1				
PD1-24A2	P161-184A1				
PD65-88A2	P225-248A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A	A	A	A
		-	B	B	B
		-	-	C	C
		-	-	D	D
		-	-	-	E
		-	-	-	F
		-	-	-	G
		-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242
24	33-56
	97-120
	161-184
	225-248

11. 8 Tester Channel Allocation Table (7)

Table 11-7 PIN MODE = A 1 Child: 8

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1	P33-56A1	8	16	32	64
PD65-88A1	P97-120A1				
PD1-24A2	P161-184A1				
PD65-88A2	P225-248A1				
PD1-24A3	P33-56A3				
PD65-88A3	P97-120A3				
PD1-24A4	P161-184A3				
PD65-88A4	P225-248A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A - - - - - - -	A B - - - - - -	A B C D - - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242
24	33-56
	97-120
	161-184
	225-248

11. 9 Tester Channel Allocation Table (8)

Table 11-8 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1	P33-40A1, P49A1, P51A1, P53A1, P55A1	128	-----	-----	-----
PD1-24A1	P41-48A1, P50A1, P52A1, P54A1, P56A1				
PD65-88A1	P97-104A1, P113A1, P115A1, P117A1, P119A1				
PD65-88A1	P105-112A1, P114A1, P116A1, P118A1, P120A1				
PD1-24A2	P161-168A1, P177A1, P179A1, P181A1, P183A1				
PD1-24A2	P169-176A1, P178A1, P180A1, P182A1, P184A1				
PD65-88A2	P225-232A1, P241A1, P243A1, P245A1, P247A1				
PD65-88A2	P233-240A1, P242A1, P244A1, P246A1, P248A1				
PD1-24A3	P33-40A3, P49A3, P51A3, P53A3, P55A3				
PD1-24A3	P41-48A3, P50A3, P52A3, P54A3, P56A3				
PD65-88A3	P97-104A3, P113A3, P115A3, P117A3, P119A3				
PD65-88A3	P105-112A3, P114A3, P116A3, P118A3, P120A3				
PD1-24A4	P161-168A3, P177A3, P179A3, P181A3, P183A3				
PD1-24A4	P169-176A3, P178A3, P180A3, P182A3, P184A3				
PD65-88A4	P225-232A3, P241A3, P243A3, P245A3, P247A3				
PD65-88A4	P233-240A3, P242A3, P244A3, P246A3, P248A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
	161
	169
	225
	233
4	33-36
	41-44
	97-100
	105-108
	161-164
	169-172
	225-228
	233-236
8	33-40
	41-48
	97-104
	105-112
	161-168
	169-176
	225-232
	233-240
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
	161-168, 177
	169-176, 178
	225-232, 241
	233-240, 242
12	33-40, 49, 51, 53, 55
	41-48, 50, 52, 54, 56
	97-104, 113, 115, 117, 119
	105-112, 114, 116, 118, 120
	161-168, 177, 179, 181, 183
	169-176, 178, 180, 182, 184
	225-232, 241, 243, 245, 247
	233-240, 242, 244, 246, 248

11.10 Tester Channel Allocation Table (9)

Table 11-9 PIN MODE = B 1 Child: 16

Driver channel	I/O channel	Number of DUTs for parallel test			
PD1-24A1	P33-41A1, P51-53A1	128	-----	-----	-----
PD1-24A1	P42-50A1, P54-56A1				
PD65-88A1	P97-105A1, P115-117A1				
PD65-88A1	P106-114A1, P118-120A1				
PD1-24A2	P161-169A1, P179-181A1				
PD1-24A2	P170-178A1, P182-184A1				
PD65-88A2	P225-233A1, P243-245A1				
PD65-88A2	P234-242A1, P246-248A1				
PD1-24A3	P33-41A3, P51-53A3				
PD1-24A3	P42-50A3, P54-56A3				
PD65-88A3	P97-105A3, P115-117A3				
PD65-88A3	P106-114A3, P118-120A3				
PD1-24A4	P161-169A3, P179-181A3				
PD1-24A4	P170-178A3, P182-184A3				
PD65-88A4	P225-233A3, P243-245A3				
PD65-88A4	P234-242A3, P246-248A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)		A B C D E F G H	-----	-----	-----

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
12	33-41, 51-53
	42-50, 54-56
	97-105, 115-117
	106-114, 118-120
	161-169, 179-181
	170-178, 182-184
	225-233, 243-245
	234-242, 246-248

11.11 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 11-10 DC Measuring Unit Allocation Table

Child	Tester channel					
	1-24	65-88	33-56	97-120	161-184	225-248
A1	DC1	DC9	DC1	DC9	DC2	DC10
A2	DC2	DC10	-----			
A3	DC33	DC41	DC33	DC41	DC34	DC42
A4	DC34	DC42	-----			
B1	DC3	DC11	DC3	DC11	DC4	DC12
B2	DC4	DC12	-----			
B3	DC35	DC43	DC35	DC43	DC36	DC44
B4	DC36	DC44	-----			
C1	DC5	DC13	DC5	DC13	DC6	DC14
C2	DC6	DC14	-----			
C3	DC37	DC45	DC37	DC45	DC38	DC46
C4	DC38	DC46	-----			
D1	DC7	DC15	DC7	DC15	DC8	DC16
D2	DC8	DC16	-----			
D3	DC39	DC47	DC39	DC47	DC40	DC48
D4	DC40	DC48	-----			
E1	DC17	DC25	DC17	DC25	DC18	DC26
E2	DC18	DC26	-----			
E3	DC49	DC57	DC49	DC57	DC50	DC58
E4	DC50	DC58	-----			
F1	DC19	DC27	DC19	DC27	DC20	DC28
F2	DC20	DC28	-----			
F3	DC51	DC59	DC51	DC59	DC52	DC60
F4	DC52	DC60	-----			
G1	DC21	DC29	DC21	DC29	DC22	DC30
G2	DC22	DC30	-----			
G3	DC53	DC61	DC53	DC61	DC54	DC62
G4	DC54	DC62	-----			
H1	DC23	DC31	DC23	DC31	DC24	DC32
H2	DC24	DC32	-----			
H3	DC55	DC63	DC55	DC63	DC56	DC64
H4	DC56	DC64	-----			

12. 3072CH Test Head with High-speed Clock Drivers

This chapter shows the tester channel allocation table used for parallel testing of multiple DUTs with a 3072CH test head with high-speed clock drivers.

12. 1 Test Head Configurations, Available Child Pins, and Channel Numbers

3072CH test head with high-speed clock drivers has three configurations, Full, Half, and Quarter, and the child pins and channel numbers shown in the table below can be used.

Test head configuration		Child		Channel number
3072CH with high-speed clock drivers	Full	A1, A2, A3, A4	B1, B2, B3, B4	1 to 248
		C1, C2, C3, C4	D1, D2, D3, D4	
		E1, E2, E3, E4	F1, F2, F3, F4	
		G1, G2, G3, G4	H1, H2, H3, H4	
	Half	A1, A2, -, -	B1, B2, -, -	1 to 248
		C1, C2, -, -	D1, D2, -, -	
		E1, E2, -, -	F1, F2, -, -	
		G1, G2, -, -	H1, H2, -, -	
	Quarter	A1, -, -, -	B1, -, -, -	1 to 120
		C1, -, -, -	D1, -, -, -	
		E1, -, -, -	F1, -, -, -	
		G1, -, -, -	H1, -, -, -	

Limitation

Fail data from the following I/O channels are not imported into fail analysis memory.

51 to 54, 115 to 118, 179 to 182, 243 to 246

12. 2 High-speed Clock Driver Channel

The high-speed clock driver channels in the table below are provided.

Channel type	Channel
High-speed clock driver channel	55-56, 119-120, 183-184, 247-248

12. 3 Tester Channel Allocation Table (1)

Table 12-1 PIN MODE = A 1 Child: 1

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1 PD65-85A1	PD55-56A1 PD119-120A1 PD183-184A1 PD247-248A1	P33-54A1 P97-118A1 P161-182A1 P225-246A1	1	2	4	8
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A - - - - - -	A B - - - - -	A B C D - - -	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
24	33-54, 55-56 ^{*1}
32	33-48, 97-112
36	33-50, 97-114
48	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1}
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242
96	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1} , 161-182, 183-184 ^{*1} , 225-246, 247-248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.



Limitation

The number of bits that exceeds 48 cannot be used in a Quarter configuration test head.

12. 4 Tester Channel Allocation Table (2)

Table 12-2 PIN MODE = A 1 Child: 2

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1 PD65-85A1	PD55-56A1 PD119-120A1 PD183-184A1 PD247-248A1	P33-54A1 P97-118A1 P161-182A1 P225-246A1	2	4	8	16
PD1-21A3 PD65-85A3	PD55-56A3 PD119-120A3 PD183-184A3 PD247-248A3	P33-54A3 P97-118A3 P161-182A3 P225-246A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A	A	A	A
			-	B	B	B
			-	-	C	C
			-	-	D	D
			-	-	-	E
			-	-	-	F
			-	-	-	G
			-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
4	33-36
8	33-40
9	33-40, 49
16	33-48
18	33-50
24	33-54, 55-56 ^{*1}
32	33-48, 97-112
36	33-50, 97-114
48	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1}
64	33-48, 97-112, 161-176, 225-240
72	33-50, 97-114, 161-178, 225-242
96	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1} , 161-182, 183-184 ^{*1} , 225-246, 247-248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.



Limitation

The number of bits that exceeds 48 cannot be used in a Quarter configuration test head.

12. 5 Tester Channel Allocation Table (3)

Table 12-3 PIN MODE = A 1 Child: 2

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1 PD65-85A1	PD55-56A1 PD119-120A1	P33-54A1 P97-118A1	2	4	8	16
PD1-21A2 PD65-85A2	PD183-184A1 PD247-248A1	P161-182A1 P225-246A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A	A	A	A
			-	B	B	B
			-	-	C	C
			-	-	D	D
			-	-	-	E
			-	-	-	F
			-	-	-	G
			-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
24	33-54, 55-56 ^{*1}
	161-182, 183-184 ^{*1}
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242
48	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1}
	161-182, 183-184 ^{*1} , 225-246, 247-248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.

12. 6 Tester Channel Allocation Table (4)

Table 12-4 PIN MODE = A 1 Child: 2

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1	PD55-56A1	P33-54A1	2	4	8	16
PD65-85A1	PD119-120A1	P97-118A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A	A	A	A
			-	B	B	B
			-	-	C	C
			-	-	D	D
			-	-	-	E
			-	-	-	F
			-	-	-	G
			-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
4	33-36
	97-100
8	33-40
	97-104
9	33-40, 49
	97-104, 113
16	33-48
	97-112
18	33-50
	97-114
24	33-54, 55-56 ^{*1}
	97-118, 119-120 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.

12. 7 Tester Channel Allocation Table (5)

Table 12-5 PIN MODE = A 1 Child: 4

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1 PD65-85A1	PD55-56A1 PD119-120A1	P33-54A1 P97-118A1	4	8	16	32
PD1-21A2 PD65-85A2	PD183-184A1 PD247-248A1	P161-182A1 P225-246A1				
PD1-21A3 PD65-85A3	PD55-56A3 PD119-120A3	P33-54A3 P97-118A3				
PD1-21A4 PD65-85A4	PD183-184A3 PD247-248A3	P161-182A3 P225-246A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A - - - - - -	A B - - - - -	A B C D E F G H	A B C D E F G H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	161
4	33-36
	161-164
8	33-40
	161-168
9	33-40, 49
	161-168, 177
16	33-48
	161-176
18	33-50
	161-178
24	33-54, 55-56 ^{*1}
	161-182, 183-184 ^{*1}
32	33-48, 97-112
	161-176, 225-240
36	33-50, 97-114
	161-178, 225-242
48	33-54, 55-56 ^{*1} , 97-118, 119-120 ^{*1}
	161-182, 183-184 ^{*1} , 225-246, 247-248 ^{*1}

**1 Specify high-speed clock channels with P. Do not specify them with PD or PC.*

12. 8 Tester Channel Allocation Table (6)

Table 12-6 PIN MODE = A 1 Child: 4

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1	PD55-56A1	P33-54A1	4	8	16	32
PD65-85A1	PD119-120A1	P97-118A1				
PD1-21A2	PD183-184A1	P161-182A1				
PD65-85A2	PD247-248A1	P225-246A1				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A	A	A	A
			-	B	B	B
			-	-	C	C
			-	-	D	D
			-	-	-	E
			-	-	-	F
			-	-	-	G
			-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242
24	33-54, 55-56 ^{*1}
	97-118, 119-120 ^{*1}
	161-182, 183-184 ^{*1}
	225-246, 247-248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.

12. 9 Tester Channel Allocation Table (7)

Table 12-7 PIN MODE = A 1 Child: 8

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1	PD55-56A1	P33-54A1	8	16	32	64
PD65-85A1	PD119-120A1	P97-118A1				
PD1-21A2	PD183-184A1	P161-182A1				
PD65-85A2	PD247-248A1	P225-246A1				
PD1-21A3	PD55-56A3	P33-54A3				
PD65-85A3	PD119-120A3	P97-118A3				
PD1-21A4	PD183-184A3	P161-182A3				
PD65-85A4	PD247-248A3	P225-246A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A	A	A	A
			-	B	B	B
			-	-	C	C
			-	-	D	D
			-	-	-	E
			-	-	-	F
			-	-	-	G
			-	-	-	H

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	97
	161
	225
4	33-36
	97-100
	161-164
	225-228
8	33-40
	97-104
	161-168
	225-232
9	33-40, 49
	97-104, 113
	161-168, 177
	225-232, 241
16	33-48
	97-112
	161-176
	225-240
18	33-50
	97-114
	161-178
	225-242
24	33-54, 55-56 ^{*1}
	97-118, 119-120 ^{*1}
	161-182, 183-184 ^{*1}
	225-246, 247-248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.

12.10 Tester Channel Allocation Table (8)

Table 12-8 PIN MODE = B 1 Child: 16

Driver channel		I/O channel	Number of DUTs for parallel test			
PD1-21A1	PD55A1	P33-40A1, P49A1, P51A1, P53A1	128	-----	-----	-----
PD1-21A1	PD56A1	P41-48A1, P50A1, P52A1, P54A1				
PD65-85A1	PD119A1	P97-104A1, P113A1, P115A1, P117A1				
PD65-85A1	PD120A1	P105-112A1, P114A1, P116A1, P118A1				
PD1-21A2	PD183A1	P161-168A1, P177A1, P179A1, P181A1				
PD1-21A2	PD184A1	P169-176A1, P178A1, P180A1, P182A1				
PD65-85A2	PD247A1	P225-232A1, P241A1, P243A1, P245A1				
PD65-85A2	PD248A1	P233-240A1, P242A1, P244A1, P246A1				
PD1-21A3	PD55A3	P33-40A3, P49A3, P51A3, P53A3				
PD1-21A3	PD56A3	P41-48A3, P50A3, P52A3, P54A3				
PD65-85A3	PD119A3	P97-104A3, P113A3, P115A3, P117A3				
PD65-85A3	PD120A3	P105-112A3, P114A3, P116A3, P118A3				
PD1-21A4	PD183A3	P161-168A3, P177A3, P179A3, P181A3				
PD1-21A4	PD184A3	P169-176A3, P178A3, P180A3, P182A3				
PD65-85A4	PD247A3	P225-232A3, P241A3, P243A3, P245A3				
PD65-85A4	PD248A3	P233-240A3, P242A3, P244A3, P246A3				
Child names to be used (When child B to H are used, their driver channel numbers and I/O channel numbers are the same as those for child A.)			A B C D E F G H	-----	-----	-----

12. 10 Tester Channel Allocation Table (8)

Bit specification (BIT=a) and corresponding I/O channels

Bit (BIT=a)	Available I/O channel
1	33
	41
	97
	105
	161
	169
	225
	233
4	33-36
	41-44
	97-100
	105-108
	161-164
	169-172
	225-228
	233-236
8	33-40
	41-48
	97-104
	105-112
	161-168
	169-176
	225-232
	233-240
9	33-40, 49
	41-48, 50
	97-104, 113
	105-112, 114
	161-168, 177
	169-176, 178
	225-232, 241
	233-240, 242
12	33-40, 49, 51, 53, 55 ^{*1}
	41-48, 50, 52, 54, 56 ^{*1}
	97-104, 113, 115, 117, 119 ^{*1}
	105-112, 114, 116, 118, 120 ^{*1}
	161-168, 177, 179, 181, 183 ^{*1}
	169-176, 178, 180, 182, 184 ^{*1}
	225-232, 241, 243, 245, 247 ^{*1}
	233-240, 242, 244, 246, 248 ^{*1}

^{*1} Specify high-speed clock channels with P. Do not specify them with PD or PC.

12.11 DC Measuring Unit Allocation Table

DC measuring units connected to respective tester channels are as shown in the table below.

Table 12-9 DC Measuring Unit Allocation Table

Child	Tester channel					
	1-21	65-85	33-56	97-120	161-184	225-248
A1	DC1	DC9	DC1	DC9	DC2	DC10
A2	DC2	DC10	-----			
A3	DC33	DC41	DC33	DC41	DC34	DC42
A4	DC34	DC42	-----			
B1	DC3	DC11	DC3	DC11	DC4	DC12
B2	DC4	DC12	-----			
B3	DC35	DC43	DC35	DC43	DC36	DC44
B4	DC36	DC44	-----			
C1	DC5	DC13	DC5	DC13	DC6	DC14
C2	DC6	DC14	-----			
C3	DC37	DC45	DC37	DC45	DC38	DC46
C4	DC38	DC46	-----			
D1	DC7	DC15	DC7	DC15	DC8	DC16
D2	DC8	DC16	-----			
D3	DC39	DC47	DC39	DC47	DC40	DC48
D4	DC40	DC48	-----			
E1	DC17	DC25	DC17	DC25	DC18	DC26
E2	DC18	DC26	-----			
E3	DC49	DC57	DC49	DC57	DC50	DC58
E4	DC50	DC58	-----			
F1	DC19	DC27	DC19	DC27	DC20	DC28
F2	DC20	DC28	-----			
F3	DC51	DC59	DC51	DC59	DC52	DC60
F4	DC52	DC60	-----			
G1	DC21	DC29	DC21	DC29	DC22	DC30
G2	DC22	DC30	-----			
G3	DC53	DC61	DC53	DC61	DC54	DC62
G4	DC54	DC62	-----			
H1	DC23	DC31	DC23	DC31	DC24	DC32
H2	DC24	DC32	-----			
H3	DC55	DC63	DC55	DC63	DC56	DC64
H4	DC56	DC64	-----			

Appendix 1. Compilation Messages, Compilation Error Codes

This appendix explains compilation messages displayed when socket programs are compiled, and compilation error codes displayed when compilation errors occurred.

A1. 1 Compilation Messages

When compilation terminated normally, the following messages are displayed on VKT.

Figure A 1-1 Compilation Messages Displayed

SOCKET S32D COMPILATION OK!	...	(1)
DR PIN MAX	:XX	... (2)
CP PIN MAX	:XX	... (3)
SOURCE LINE NUMBER	:XX	... (4)
OBJECT FILE SIZE (BYTE)	:XX	... (5)
COMPILATION TIME (SEC)	:XX	... (6)
LINE/MINUTE	:XX	... (7)

- (1) Socket program S32D is compiled normally.
- (2) DR pin MAX value
- (3) CP pin MAX pin value
- (4) Number of lines in the source file (socket program). (in decimal number)
- (5) Size of the object file (in bytes in decimal number)
- (6) Time taken for compilation (in seconds)
- (7) Number of lines compiled in a minute (in decimal number)

A1. 2 Compilation Error Codes

The output format of compilation error codes is the same as the one for test plan programs. Explanations of error codes related to socket programs are as follows:

Error code	Description
1600	The program does not begin with SOCKET.
1601	A meaningless character is included. (Example) A0-3 = PD5, _ PD6, PD7
1602	The specified value exceeds the limit value.
1603	A registered character is used in the pin name. (Example) BIT = PD2
1604	A numeric character is included between alphabetical characters. (Example) A3XB120
1605	The same item is already specified in the line.
1606	The item in a different format is specified in the line.
1607	The start and end values are the same in consecutive specification (mm-nn). (Example) MA10-10 = ...
1610	The necessary item is not specified in the line. (Example) WE = PD3 = ... (Tester channel number is not specified.)
1611	The necessary registered character is not specified in the line. (Example) DEFINE ____ (The necessary character string is not specified.)
1612	More than 96 user symbols are specified.
1613	The number of DUTs and stations are not specified.
1614	The number of bits is not specified.
1615	The number of bits specified is out of range. (Example) BIT = 3
1616	The number of PPSs or VSs is not specified.
1617	The number of IOPINs or DRPINs is not specified.
1620	* does not exist in IOPIN or DRPIN specification. (Example) DRPIN = _4DR
1621	BIT description does not exist.

Error code	Description
1622	PPS or VS description does not exist.
1623	DRPIN description does not exist.
1624	IOPIN description does not exist.
1625	DUT# description does not exist.
1626	The numbers of pins of DUT# and DUTn do not agree. (Example) DUT# = VS1, PD1-3 DUT1 = PPS3, PD1-4C1
1627	The DUT number on the tester is not decided.
1631	Pin numbers exceeding 44, the maximum value of pin numbers for each of which two or more PIN numbers can be specified, are specified.
1632	All tester channels allocated to one DUT must be connectable to the same test unit.
1633	The DUT number on the tester exceeds the total number of DUTs.
1634	The numbers of BITs and comparator channels do not agree.
1635	Unavailable tester channel number is specified.
1636	Tester channels having the same child name (A, B, C, D, E, F, G, or H) must be allocated for one DUT when specifying 1 to 24 or 33 to 52 pins. (Example) DUT1 = PC33A, PC34B ← error
1640	In parallel test definition, PPSn and PPSn+16 do not belong to respective PPS groups of DUT#. (n = 1 ~ 12) (Example) DUT# = PPS1, PPS2 DUT1 = PPS1, PPS5 DUT2 = PPS2, PPS17 ← error
1641	In parallel test definition, a tester channel that cannot be allocated to the DUT is specified. (Example) DUT# = PPS1, PD1, P33 DUT1 = PPS2, PD1A1, P33B ← error
1642	The value in the PPS statement does not agree with the number of PPSs in DUT# specification. (Example) PPS = 3DUT# = VS1, PD1, P33 ← error (because VS2 and VS3 do not exist) DUT1 = PPS2, PD1A1, P33A
1643	For DUT#, PPS numbers are not specified in ascending order. (Example) DUT# = PPS1, PPS3, PPS2 ← error DUT1 = PPS1, PPS4, PPS17

A1. 2 Compilation Error Codes

Error code	Description
1644	Between DUT# and DUTn, the driver channel or I/O channel does not correspond to the DR pin side or the CP pin side, respectively. (Example) DUT#=VS1, PD1, P33 DUT1=PPS2, PD1A1, <u>P2A1</u> ← error
1645	Values different from PPS, PD, P, and PC of DUT# are defined for DUTn. (Example) DUT# = VS1, PD1, P33 DUT1 = PPS2, P33A, PD1A1 ← error
1646	In parallel test definition, the number of DUTs specified and the maximum value of DUT numbers specified do not agree.
1647	In the specified PIN MODE, the number of DUTs specified is out of range.
1650	The pin numbers of DUT# and DUTn do not agree. (Example) DUT# = PPS1, PD1, P33, <u>PD34</u> DUT1 = PPS1, PD1A1, P33A, PD34A DUT2 = PPS2, PD1B1, P33B, <u>PD35B</u> ← error
1651	There is an error in the specification of the driver channel number.
1652	In parallel test definition, the minimum pin number in one DUT must be either of 1, 11, 65, 75, 129, 193, 257, 321, 385, and 449.
1653	There is an error in the specification of the I/O channel number.
1654	Pin numbers specified do not correspond to the number of bits specified.
1655	The driver channel is not specified for DUT#.
1656	In parallel test definition, when a common child name is used in two DUTn specifications, both PD1 and PD13 are specified in one DUT. (Example) DUT# = VS1, PD1, PD2, P33 DUT1 = VS1, <u>PD1A1</u> , <u>PD13A1</u> , P33A ← error DUT2 = VS2, PD14A1, PD15A1, P41A
1657	In parallel test definition, the total number of tester channels or VSs exceeds its maximum number.
1660	The number of pin definition tables exceeds 65.
1661	Socket file name is not specified. Or it is not correctly specified.
1662	In HANDLER or PROBER specification, the file name is not written.
1663	The PIN MODE is not specified.
1664	The number of user symbols exceeds the limit value.
1665	Definitions for parallel test not allowed in the specified PIN MODE are specified.
1667	FMALL (FMALL1) can not be specified.
1670	There is an error in the combination of the CHTYPE and PIN MODE.

Error code	Description
1672	In HVPPS specification, the channel number on the left side must be the same as the minimum channel number of the channel numbers specified on the right side.
1673	In parallel test definition, the numbers of PPSs of DUT# and DUTn do not agree.
1674	In parallel test definition, the pin number specified is already used.
1675	The pin number of DUT# is not the minimum pin number of DUTn.
1676	The pin number of DUT# is not specified for DUTn.
1677	You cannot specify VS5 and VS9, VS6 and VS10, VS7 and VS11, or VS8 and VS12 simultaneously.

Appendix 2. Example of Socket Program

This appendix shows an example of a socket program description.

A2. 1 Example of Socket Program Description

◆ Source List

An example of parallel test of 16 DUTs in two stations when PIN MODE = A and 1 bit.

```

00001      SOCKET  S32D   16      16
00002          A0-7 = PD5, PD7, PD6, PD12, PD11, PD10, PD13, PD9 = 1-8
00003          WE   = PD3                                = 9
00004          RAS  = PD4                                = 10
00005          CAS  = PD15                               = 11
00006          DIN  = PD2                                = 33
00007          DOUT = PC14                               = 33
00008          RFSB = PD1                                = 12
00009      ;
00010      DEFINE  SYMBOL
00011          VCC   = VS1                                ;; P8
00012          VSS   = GND                                ;; P16
00013      ;
00014      DEFINE  MULTIPLE      8DUT  2STN
00015          BIT      = 1
00016          PPS      = 1
00017          PIN MODE = A
00018          CHTYPE4
00019          HANDLER
00020
00021          DUT# = VS1, PD1-12, PD34, PC33
00022          DUT1 = PPS1, PD1-12A1, PD34A, PC33A
00023          DUT2 = PPS2, PD1-12B1, PD34B, PC33B
00024          DUT3 = PPS3, PD1-12C1, PD34C, PC33C
00025          DUT4 = PPS4, PD1-12D1, PD34D, PC33D
00026          DUT5 = PPS7, PD1-12E1, PD34E, PC33E
00027          DUT6 = PPS8, PD1-12F1, PD34F, PC33F
00028          DUT7 = PPS9, PD1-12G1, PD34G, PC33G
00029          DUT8 = PPS10, PD1-12H1, PD34H, PC33H
00030      END

```

◆ **Compilation Result**

PAGE 1 SOC 32D

* ASSIGNMENT OF DRIVER PIN

DUT PIN #	TESTER CH
PD1	12
PD2	33
PD3	9
PD4	10
PD5	1
PD6	3
PD7	2
PD8	13
PD9	8
PD10	6
PD11	5
PD12	4
PD13	7
PD14	14
PD15	11
PD16	15

* ASSIGNMENT OF COMPARATOR PIN

DUT PIN #	TESTER CH
PC1	1
PC2	2
PC3	3
PC4	4
PC5	5
PC6	6
PC7	7
PC8	8
PC9	9
PC10	10
PC11	11
PC12	12
PC13	13
PC14	33
PC15	15
PC16	16

DEFINED NAME RESERVED NAME

A0	PD5
A1	PD7
A2	PD6
A3	PD12
A4	PD11
A5	PD10
A6	PD13
A7	PD9
WE	PD3
RAS	PD4
CAS	PD15
DIN	PD2
DOUT	PC14
RFSH	PD1
VCC	VS1
VSS	GND

List of Figures

Figure 2-1	Configuration of Socket Program	2-1
Figure 2-2	Definition of DUT Pin Name	2-4
Figure 2-3	Consecutive Specification of DUT Pin Names	2-5
Figure 2-4	Definition of Converted Symbols	2-9
Figure 2-5	Format of Definition for Parallel Test	2-12
Figure 2-6	Changes in the Voltage of VS9, VS10, VS11 and VS12	2-15
Figure 2-7	Consecutive Specification of Channel Numbers	2-24
Figure 3-1	Structure of Tester Channel Allocation Table	3-1
Figure A 1-1	Compilation Messages Displayed	A1-1

List of Tables

Table C-1	Applicable Models to Descriptions of Test System	Convention-1
Table C-2	Test Head Corresponding to Test System	Convention-2
Table C-3	Reference Indicator	Convention-2
Table 2-1	Available Formats of DUT Pin Numbers	2-7
Table 2-2	Available Format	2-8
Table 2-3	CHTYPE _n and Available Bit Numbers	2-13
Table 2-4	CHTYPE _n and Channel Number Type of Test Head	2-13
Table 4-1	PIN MODE = A 1 Child: 1	4-3
Table 4-2	PIN MODE = A 1 Child: 2	4-4
Table 4-3	PIN MODE = A 1 Child: 4	4-5
Table 4-4	PIN MODE = B 1 Child: 8	4-7
Table 4-5	PIN MODE = A 1 Child: 2	4-9
Table 4-6	PIN MODE = A 1 Child: 2	4-10
Table 4-7	PIN MODE = B 1 Child: 8	4-11
Table 4-8	DC Measuring Unit Allocation Table (1)	4-13
Table 4-9	DC Measuring Unit Allocation Table (2)	4-14
Table 5-1	PIN MODE = A 1 Child: 1	5-3
Table 5-2	PIN MODE = A 1 Child: 2	5-4
Table 5-3	PIN MODE = B 1 Child: 4	5-5
Table 5-4	PIN MODE = A 1 Child: 2	5-6
Table 5-5	DC Measuring Unit Allocation Table (1)	5-7
Table 5-6	DC Measuring Unit Allocation Table (2)	5-8
Table 5-7	DC Measuring Unit Allocation Table (3)	5-9
Table 6-1	PIN MODE = A 1 Child: 1	6-5
Table 6-2	PIN MODE = A 1 Child: 2	6-6
Table 6-3	PIN MODE = A 1 Child: 2	6-7
Table 6-4	PIN MODE = A 1 Child: 4	6-8
Table 6-5	PIN MODE = B 1 Child: 4	6-9
Table 6-6	PIN MODE = B 1 Child: 8	6-10
Table 6-7	PIN MODE = B 1 Child: 8	6-11
Table 6-8	PIN MODE = C 1 Child: 16	6-13
Table 6-9	DC Measuring Unit Allocation Table (1)	6-15
Table 6-10	DC Measuring Unit Allocation Table (2)	6-16
Table 6-11	DC Measuring Unit Allocation Table (3)	6-17
Table 7-1	PIN MODE = A 1 Child: 1	7-2

Table 7-2	PIN MODE = A 1 Child: 2	7-3
Table 7-3	PIN MODE = A 1 Child: 2/4	7-5
Table 7-4	PIN MODE = A 1 Child: 2/4/8	7-7
Table 7-5	PIN MODE = B 1 Child: 2/4/8/16	7-9
Table 7-6	PIN MODE = A 1 Child: 1/2/4/8	7-12
Table 7-7	PIN MODE = B 1 Child: 2/4/8/16	7-14
Table 7-8	DC Measuring Unit Allocation Table	7-17
Table 8-1	PIN MODE = A 1 Child: 2	8-3
Table 8-2	PIN MODE = A 1 Child: 4	8-4
Table 8-3	PIN MODE = A 1 Child: 8	8-6
Table 8-4	PIN MODE = B 1 Child: 16	8-8
Table 8-5	PIN MODE = B 1 Child: 16	8-10
Table 8-6	DC Measuring Unit Allocation Table (1)	8-12
Table 8-7	DC Measuring Unit Allocation Table (2)	8-13
Table 9-1	PIN MODE = A 1 Child: 1	9-2
Table 9-2	PIN MODE = A 1 Child: 2	9-3
Table 9-3	PIN MODE = A 1 Child: 4	9-4
Table 9-4	PIN MODE = A 1 Child: 2	9-6
Table 9-5	DC Measuring Unit Allocation Table (1)	9-7
Table 9-6	DC Measuring Unit Allocation Table (2)	9-7
Table 9-7	DC Measuring Unit Allocation Table (3)	9-8
Table 9-8	DC Measuring Unit Allocation Table (4)	9-8
Table 10-1	PIN MODE = A 1 Child: 1	10-3
Table 10-2	PIN MODE = A 1 Child: 2	10-4
Table 10-3	PIN MODE = A 1 Child: 2	10-5
Table 10-4	PIN MODE = A 1 Child: 2	10-6
Table 10-5	PIN MODE = A 1 Child: 4	10-7
Table 10-6	PIN MODE = A 1 Child: 4	10-8
Table 10-7	PIN MODE = A 1 Child: 4	10-9
Table 10-8	PIN MODE = A 1 Child: 8	10-10
Table 10-9	PIN MODE = B 1 Child: 8	10-11
Table 10-10	PIN MODE = B 1 Child: 16	10-13
Table 10-11	PIN MODE = B 1 Child: 16	10-15
Table 10-12	PIN MODE = B 1 Child: 8	10-17
Table 10-13	PIN MODE = C 1 Child: 32	10-19
Table 10-14	PIN MODE = C 1 Child: 16	10-21
Table 10-15	PIN MODE = A 1 Child: 2	10-23
Table 10-16	PIN MODE = A 1 Child: 4	10-24

Table 10-17 PIN MODE = B 1 Child: 8	10-25
Table 10-18 PIN MODE = B 1 Child: 8	10-26
Table 10-19 DC Measuring Unit Allocation Table	10-27
Table 11-1 PIN MODE = A 1 Child: 1	11-2
Table 11-2 PIN MODE = A 1 Child: 2	11-3
Table 11-3 PIN MODE = A 1 Child: 2	11-4
Table 11-4 PIN MODE = A 1 Child: 2	11-5
Table 11-5 PIN MODE = A 1 Child: 4	11-6
Table 11-6 PIN MODE = A 1 Child: 4	11-7
Table 11-7 PIN MODE = A 1 Child: 8	11-9
Table 11-8 PIN MODE = B 1 Child: 16	11-11
Table 11-9 PIN MODE = B 1 Child: 16	11-13
Table 11-10 DC Measuring Unit Allocation Table	11-14
Table 12-1 PIN MODE = A 1 Child: 1	12-3
Table 12-2 PIN MODE = A 1 Child: 2	12-4
Table 12-3 PIN MODE = A 1 Child: 2	12-5
Table 12-4 PIN MODE = A 1 Child: 2	12-6
Table 12-5 PIN MODE = A 1 Child: 4	12-7
Table 12-6 PIN MODE = A 1 Child: 4	12-9
Table 12-7 PIN MODE = A 1 Child: 8	12-11
Table 12-8 PIN MODE = B 1 Child: 16	12-13
Table 12-9 DC Measuring Unit Allocation Table	12-15

Index

[Symbols]

%REMARK..... 2-25

[D]

DEFINE MULTIPLE 2-12

DEFINE SYMBOL 2-9

DEFINE VS 2-10

[E]

END..... 2-1

[H]

HVPPS 2-14

[P]

PARALLEL VS 2-16

[S]

SOCKET 2-1